CHE 396 Mini-Project-2

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<u>Team 2</u>

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Task 1

1. Project Charter

The objective of this project was to design an energy-efficient heat exchange network for a multi-step chemical process, utilizing pinch analysis, to minimize hot and cold utility consumption. The process involved a preheated feed stream that was heated from 60℃ to 220℃ prior to entering an endothermic reactor, followed by reheating a reactor effluent from 30℃ to 180℃. Finally, a distillate was cooled from 160℃ to 40℃, and the bottom products were cooled from 180℃ to 40℃. A target minimum temperature approach (ΔTmin) was set at 10℃ to balance capital investment and operating costs.

The scope of this project encompasses the entire heat integration workflow, which includes identifying hot and cold streams, developing composite and grand-composite curves, calculating pinch temperatures, optimizing ΔTmin, and final HEN targeting. To quantify energy savings, a comprehensive before-and-after utility analysis was conducted, and AI tools, including ChatGPT-5, Gemini 2.5 Pro, Claude, Perplexity, and Mistral, were utilized for comparison with manual calculations to assess the accuracy, assumptions, and reliability of AI-assisted engineering design.

Key results included a calculated pinch temperature of 155℃, which corresponded to a hot pint of 160℃ and a cold pinch of 150℃. The minimum hot utility was calculated to be 2.36 MW, and a minimum cold utility of 1.86 MW. The energy savings were estimated to be approximately 5.74 MW for each hot and cold utility, corresponding to a 71% reduction from the baseline. Regarding the economic impact, the estimated annual savings were determined to be approximately $1.6 million, with a payback period of less than six months.

The deliverables in this project include composite and grand composite curves, ΔTmin optimization, and $T_{pin}C_h$ calculation tables. Additionally, the project includes a heat exchanger network schematic above and below the pinch, a utility cost comparison before and after integration, an AI vs. manual results table, and a team reflection.

2. <u>Design Plan</u>

Based on the direction of the temperature change, all relevant streams were classified as either hot or cold. The feed, which went from 60°C to 220°C, and the reactor product, which went from 30°C to 180°C, were defined as cold streams. Similarly, the distillate, which was heated from 160°C to 40°C, and the bottom products, which were heated from 180°C to 40°C, were defined as the hot streams. The flow rate basis was 10 kg/s with heat capacities of 3.0, 2.2, 10.0, and 3.33 kJ/kg°C, respectively.

For each stream, the heat capacity flow (C_p x ṁ) and enthalpy change ($Q = \dot{m}C_p\Delta T$) was calculated as C_1 = 30 kW/°C (4,800 kW), C_2 = 22 kW/°C (3,300 kW), H_1 = 40 kW/°C (4,800 kW), H_2 = 20 kW/°C (2,800 kW). This resulted in a total heating demand of 8,100 kW and cooling availability of 7,600 kW.

For the temperature interval, all the streams were temperature-shifted by ±(ΔTmin/2) = ±5°C to construct the composite curves and perform the enthalpy-cascade analysis. The minimum cumulative heat deficit of -2,360 kW was used to identify the pinch temperature region at 155°C, which corresponded to a hot pinch of 160°C and a cold pinch of 150°C.

For ΔT_{min}, three values were evaluated: 5°C, 10°C, and 15°C. ΔT_{min} = 5°C yielded the maximum recovery but also resulted in a high capital cost. ΔT_{min} = 15°C minimized the area but wasted energy. Finally, ΔT_{min} = 10°C ultimately provided the optimal balance between utility savings and exchanger count, as confirmed by both manual and AI results.

To synthesize the heat exchanger network, assuming no heat transfer across the pinch, above the pinch, H2 (180°C to 160°C) heated C1 (150°C to 163.3°C, 400kW), and the remaining C1 and C2 segments were heated via a hot utility, approximately 2.36 MW in total. Below the pinch, H_1 and H_2 supplied C_1 and C_2, while approximately 5.74 MW of heat was recovered, and a residual 2.20 MW was rejected as cold utility. This network was used to achieve ΔT_{min} compliance and matched the target utility loads.

To compare the manual results with the AI output, each member entered the process into distinct AI tools. After designing the process manually and comparing it with the AI outputs, ChatGPT and Gemini produced identical pinch and utility targets to those in the manual results. However, Claude and Perplexity showed minor variations due to different cascade formulations. This exercise highlighted AI's strength in data automation and calculations, but also its limitations in thermodynamic reasoning.

3. PFDs:

Figure 2: Before Heat Integration

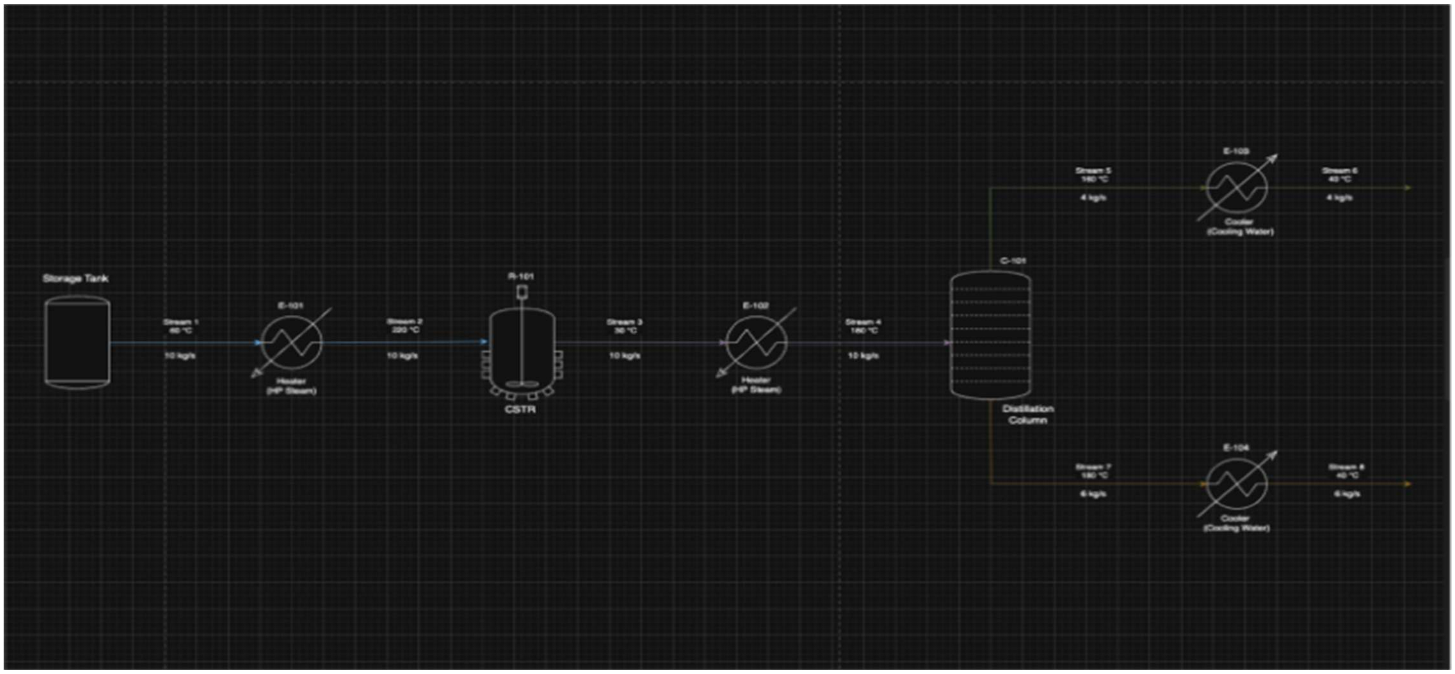
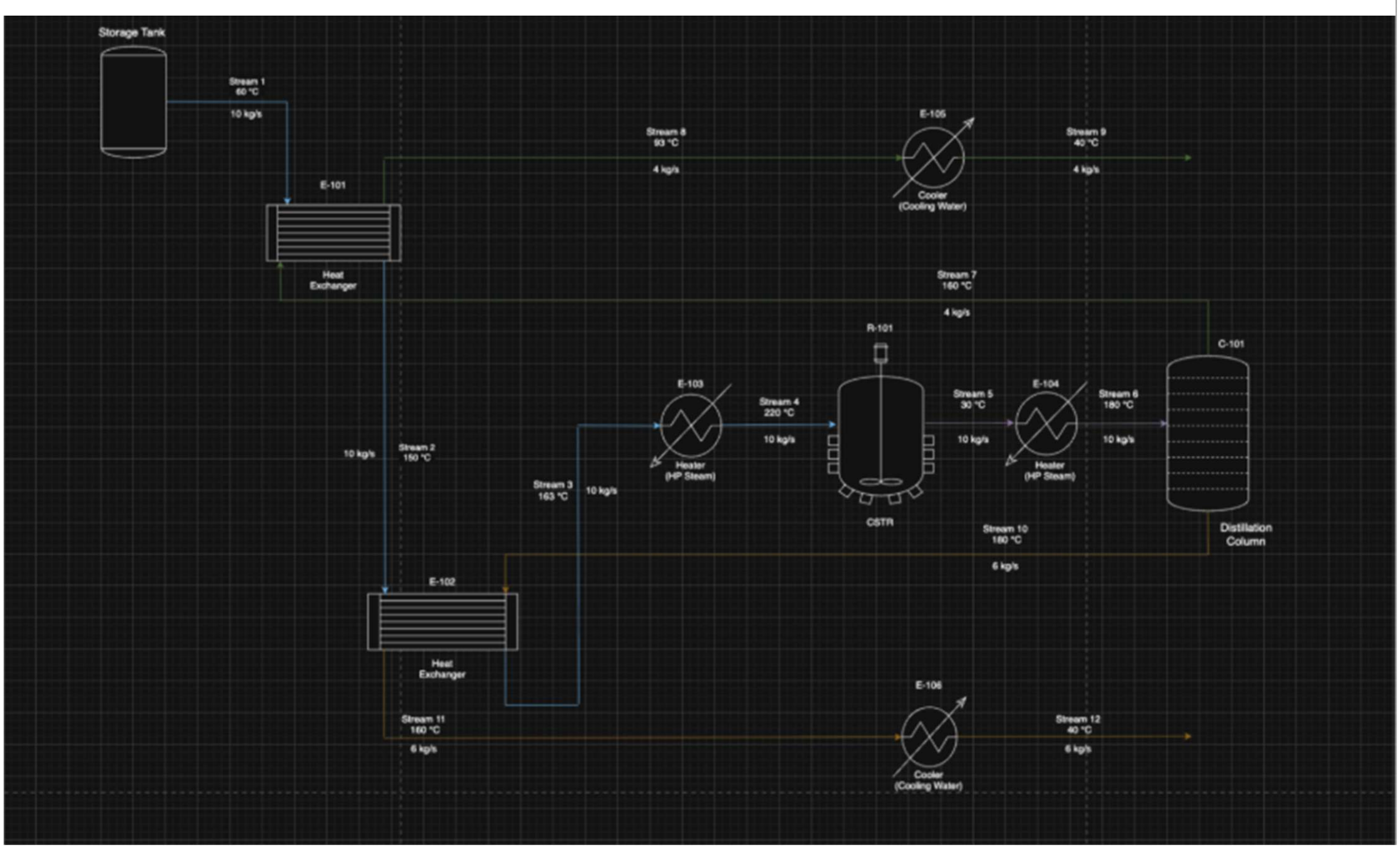


Figure 2: PFD After HEN Analysis



4. Composite Curves

Figure 3: Cold Stream Composite Curve

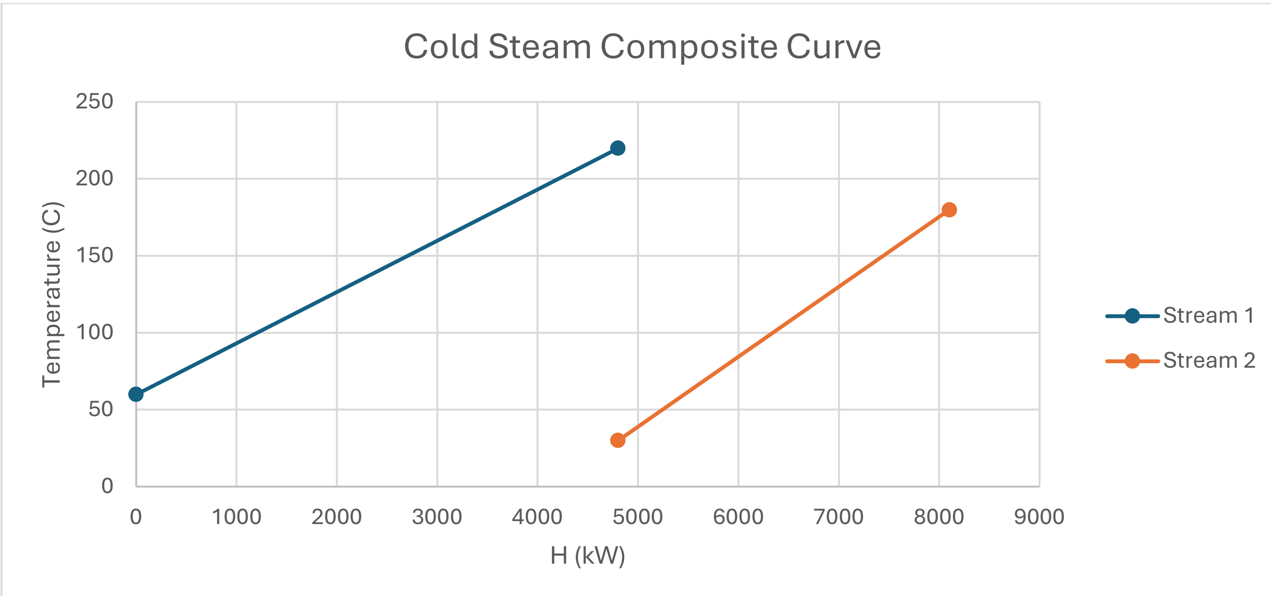
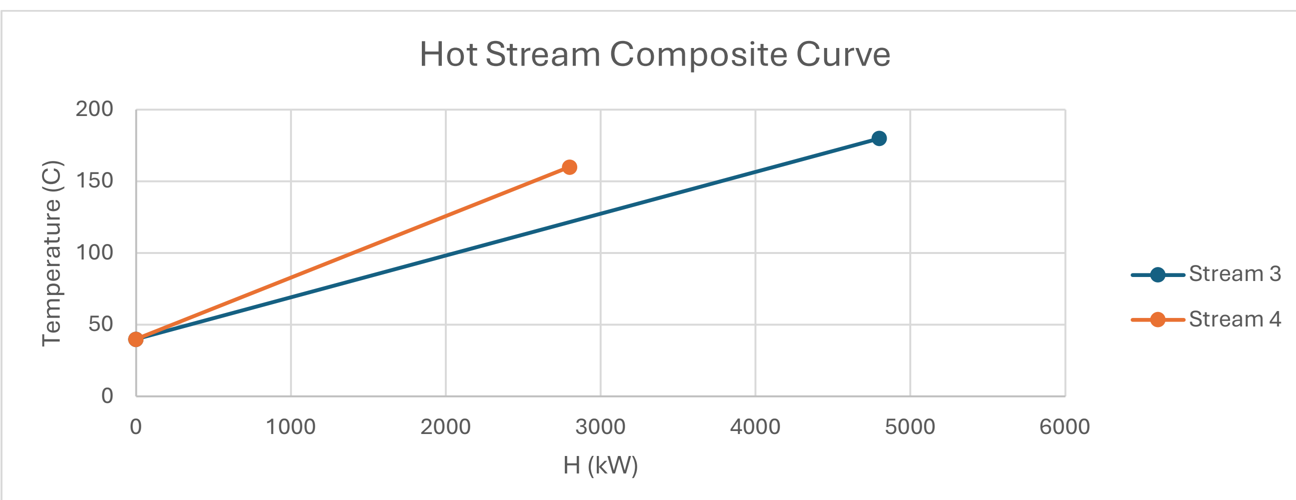


Figure 4: Hot Stream Composite Curve



To get the composite curves, plot the temperature against the enthalpy. For the cold streams, start at the lowest temperature and work your way up. For hot streams, start at the highest temperature and work your way down. To find the enthalpy, use the equation: $\Delta H = Cp\ \Delta T$. Use the corresponding streams Cp and ΔT. In terms of trends, the graphs show that the enthalpy is increasing for cold streams since they are being heated. Additionally, the enthalpy for hot streams starts high but decreases as they are cooled.

5. ΔT_{min} Calculation & Explanation

The problem states to select an optimum $\Delta T_{min} \approx 10$ °C. To make sure this is optimal, the problem is solved thrice, once using $\Delta T_{min} = 5$ °C, then $\Delta T_{min} = 10$ °C, and lastly $\Delta T_{min} = 15$ °C. In summary, setting $\Delta T_{min} = 5$ °C allows for the best heat recovery. However, this approach has a significantly larger capital cost, as the heat exchange network is larger. Meanwhile, setting $\Delta T_{min} = 15$ °C allows for a smaller heat exchange network, resulting in lower capital costs. However, there are significant heat losses. Lastly, setting $\Delta T_{min} = 10$ °C seems to be a perfect blend of optimal capital and operational costs. The heat exchange network is a reasonable size, and the heat losses are not as significant. The problem will be solved using $\Delta T_{min} = 10$ °C.

6. Pinch Temperature Calculation

To begin solving this problem, it is essential to break down the given process and label each stream as either hot or cold. Hot streams are cooled, and cold streams are heated. Additionally, a stream that undergoes a reaction or a composition change is not classified as either hot or cold.

Initially, the stream is kept at $60°\text{C}$ and heated to $220°\text{C}$, which makes it a cold stream. This is labeled as *Stream 1*. Afterward, it is sent to a reactor where an endothermic reaction takes place, and the stream leaves at $30°\text{C}$. This is neither a hot nor a cold stream since a reaction occurred. Then, the stream is heated from $30°\text{C}$ to $180°\text{C}$, making it a cold stream. This is labeled as *Stream 2*. This stream is sent to a distillation column where the distillate leaves at $160°\text{C}$ and is cooled to $40°\text{C}$, making it a hot stream. This is labeled as *Stream 3*. Note that the stream within the distillation column is not labeled as hot or cold because of the composition changes within the column. The bottom product leaves the distillation column at $180°\text{C}$ and is cooled to $40°\text{C}$, making it a hot stream. This is labeled *Stream 4*. Now that all streams are labeled, the heat capacity flow rates, in $\frac{kW}{°\text{C}}$, can be found. The problem gives heat capacities in units of $\frac{kJ}{kg\,°\text{C}}$, and the flow rate in $\frac{kg}{s}$. One kW is equivalent to one $\frac{kJ}{s}$, so by multiplying $\frac{kJ}{kg\,°\text{C}}$ and $\frac{kg}{s}$, the desired units of $\frac{kW}{°\text{C}}$ is obtained. The feed flow rate is $10\ \frac{kg}{s}$, and the heat capacity of the feed is $3\ \frac{kJ}{kg}$, so the heat capacity flow rate of stream 1 is $30\ \frac{kW}{°\text{C}}$. For stream 2, assuming steady state, the flow rate is $10\ \frac{kg}{s}$ and the heat capacity is $2.2\ \frac{kJ}{kg}$, so the heat capacity flow rate of stream 2 is $22\ \frac{kW}{°\text{C}}$. For stream 3, the problem states the distillate is 40% mass, meaning its flow rate is $4\ \frac{kg}{s}$. The heat capacity of stream 3 is $10\ \frac{kJ}{kg}$, so the heat capacity flow rate of stream 3 is $40\ \frac{kW}{°\text{C}}$. Since the distillate is 40% mass, the bottom flow rate must be 60%, making it $6\ \frac{kg}{s}$. The heat capacity of stream 4 is $3.33\ \frac{kJ}{kg}$, so the heat capacity flow rate of stream 4 is $19.98\ \frac{kW}{°\text{C}}$. Now, all the information needed to find T_{pinc} is known. Furthermore, the problem states that ΔT_{min} is 10 °C. T_{pinc} is found using Excel. Attached below are the tables.

Table 1: Inlet Values of Excel Calculated Values for T-Pinch						
	Actual temperature (ºC)		Interval temperature (ºC)		Heat capacity flow	Heat load
Stream No.	Source	Target	Source	Target	rate *CP* (kW/ºC)	(kW)
1	60	220	65	225	30	4800
2	30	180	35	185	22	3300
3	160	40	155	35	40	4800
4	180	40	175	35	19.98	2797.2

Table 2: Results of Excel Calculated Values for T-Pinch						
Interval	Interval temp (ºC)	Interval (D*T* (ºC)	Sum *CP*c - sum *CP*h (kW/ºC)	d*H* (kW)	Cascade (kW)	(kW)
	225				0	2360.4
1	185	40	30	1200	-1200	1160.4
2	175	10	52	520	-1720	640.4
3	155	20	32.02	640.4	-2360.4	0
4	65	90	-7.98	-718.2	-1642.2	718.2

T_{pinch} occurs at the lowest cascade power. According to table, T_{pinc} is 155 ℃. From here, the hot and cold pinch temperatures can be calculated. The formula to find $T_{hot,pinc}$ is:

$$T_{hot,pinc} = T_{pi} + \frac{\Delta T_{min}}{2}$$

Plug in the values:

$$T_{hot,pinc} = 155 + \frac{10}{2} = 160\ ℃$$

The formula to find $T_{cold,pinc}$ is:

$$T_{cold,pinch} = T_{pinch} - \frac{\Delta T_{min}}{2}$$

Plug in the values:

$$T_{cold,pinch} = 155 - \frac{10}{2} = 150\ ℃$$

7. Heat Exchanger Network Design

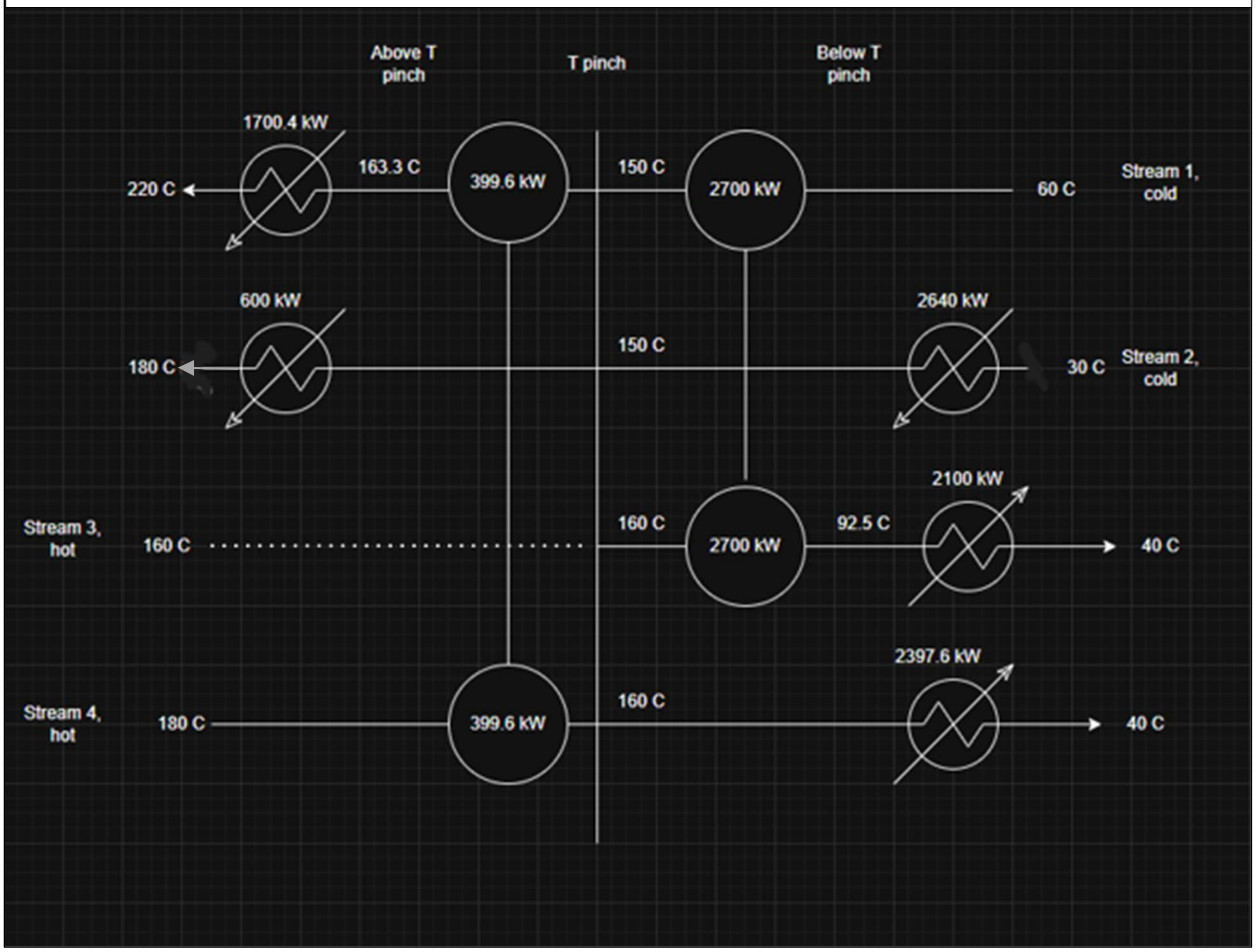
The HEN is necessary because it allows streams that require heat to be heated by streams that require cooling. This allows for operational cost savings since less external heating and cooling are required.

The process of matching streams involves looking at the Cp values. To match streams above T_{pinch}, the Cp of the cold stream should be equal to or less than the hot stream. To match streams below T_{pinch}, the Cp of the hot stream should be greater to or equal to the Cp of the cold stream.

To calculate the enthalpy of the streams, use the following equation: $\Delta H = Cp\ \Delta T$. Use the Cp and ΔT values for each respective stream.

The HEN design was made by matching the correct streams above and below the T_{pinch}. Then, the enthalpy was found for each matched stream. Then, the difference between the matched streams enthalpies was found. This was done to see if any additional heating or cooling is required for the streams that are exchanging heat. If any additional heating or cooling is needed, then either a heater or condenser is added.

Figure 5: HEN Design



8. Utility Cost Analysis

Utility Cost (Before Heat Integration)

$$Q_1 = mC_p\Delta T = (10 \text{ kg/s})(3.8\frac{kJ}{kg.°C}))(220°C - 60°C) = 4800 \text{ kW}$$

$$Q_1 = m_w C_{p,w}\Delta T \Rightarrow 4800 = m_w(4.18\frac{kJ}{kg.°C}))(220°C - 60°C)$$

$$m_w = 7.18 \text{ kg/s}$$

Cost:

$$4800 \text{ kW} \times 8000 \text{ h} = 3.84 \times 10^7 \text{ kWh}$$

$$3.84 \times 10^7 \text{ kWh} \times \$0.1286 = \$4.94 \text{ MM/yr}$$

For boiler feed water:

$$7.18 \text{ kg/s} \times 2.88 \times 10^7 \text{ s} \times \frac{2.2 \text{ lb}}{1 \text{ kg}} \times \frac{1 \text{ gal}}{8.34 \text{ lb}} \times \$3.00 = 3.834 \times 10^4 \text{ gal}$$

Boiler feed water cost = **$163,500/yr**

Boiler and Feedwater Calculations

$$Q_2 = mC_{p,fw}\Delta T = (10 \text{ kg/s})(2.2 \frac{kJ}{kg.°C}))(180°C - 30°C) = 3300 \text{ kW}$$

$$Q_2 = m_w C_{p,w}\Delta T \Rightarrow m_w = 5.26 \text{ kg/s}$$

Costs:

Boiler = **$3.4 MM/yr**
Boiler Feed Water = **$119,882.6/yr**

Cooling and Pump Work

$$Q_3 = mC_{p,IO}\Delta T = (4 \text{ kg/s})(10 \frac{kJ}{kg.°C}))(40°C - 160°C) = 4800 \text{ kW}$$

$$Q_3 = m_w C_{p,w}\Delta T \Rightarrow m_w = 9.67 \text{ kg/s}$$

Costs:
Water Pump/Fan = **$4.94 MM/yr**
Cooling Water = **$220,392.5/yr**

$$Q_4 = m_B C_{p,B}\Delta T = (6 \text{ kg/s})(3.33 \frac{kJ}{kg.°C})(40°C - 180°C) = -2797.2 \text{ kW}$$

$$Q_4 = m_w C_{p,w} \Delta T \Rightarrow m_w = 4.78 \text{ kg/s}$$

Costs:
Water Pump/Fan = **$2.9 MM/yr**
Cooling Water = **$108,842.7/yr**

Total Utility Costs (Before Heat Integration)

Power = **$16.18 MM/yr**
Water Supply = **$612,717.8/yr**

After Heat Integration

Total Power:

$$9438 \text{ kW} \times 8000 \text{ h} \times \$0.1286/\text{kWh} = \$9.71 \text{ MM/yr}$$

Total Water Supply:

$$26.6 \text{ kg/s} \times 2.88 \times 10^7 \text{ s} \times \frac{2.2 \text{ lb}}{1 \text{ kg}} \times \frac{1 \text{ gal}}{8.34 \text{ lb}} \times \$3.00 = \$605{,}411.9/\text{yr}$$

9. AI Prompt & Output Appendix

Table 3: AI Prompts and Output

Group Member	AI Version	AI Input	AI Output
Hiba Naqvi	ChatGPT 5 Thinking	The Feed stream is at 60◦ C in the storage and needs to be heated up to 220◦ C before being	Sweet—this is a perfect “pinch analysis + HEN targeting” problem. I’ll (1) define the streams, (2) do the ΔTmin=10 °C targeting (shifted temperatures, interval balances, cascade), (3) give composite/pinch results, (4) lay out a feasible heat-exchanger network (above & below pinch), and (5) show utility

		fed to the reactor. An endothermic reaction occurs in the reactor and the product leaves at. o Pinch temperature calculation o Composite curves o ΔTmin optimization o Heat exchanger network design o Utility cost calculation before and after integration 30◦ C. The reactant stream is further heated to 180◦ C and sent to a distillation column to recover the product. The liquid product from the column is at 160◦ C	loads before vs. after (you can plug in your own \$/kWh or \$/GJ later). **1) Stream data (given / inferred)** **Cold streams (need heating)** • **C1 – Feed to reactor:** 10 kg/s, Cp=3 kJ/kg °C → $\dot{C} =$ 30 kW/°C, 60 → 220 °C Heat duty: $Q_{C1} = 30 \times (220 - 60) = 4{,}800$ kW • **C2 – Reactor outlet to column (reactor product):** 10 kg/s, Cp=2.2 kJ/kg °C → $\dot{C}$ = 22 kW/°C, 30 → 180 °C Heat duty: $Q_{C2} = 22 \times (180 - 30) = 3{,}300$ kW **Hot streams (need cooling) – from distillation** • **H1 – Distillate (40% = 4 kg/s):** Cp=10 kJ/kg °C → $\dot{C} =$ 40 kW/°C, 160 → 40 °C Heat to remove: $Q_{H1} = 40 \times (160 - 40) = 4{,}800$ kW • **H2 – Bottoms (60% = 6 kg/s):** Cp=3.33 kJ/kg °C → $\dot{C} \approx 20$ kW/°C, 180 → 40 °C Heat to remove: $Q_{H2} = 20 \times (180 - 40) = 2{,}800$ kW

		and must be cooled to 40◦ C to be stored. The bottom product from the column is cooled from 180◦ C to 40◦ C. Design a heat exchanger network for this process to minimize utility consumption using pinch analysis and heat integration techniques. Target an optimum ΔTmin ≈ 10 °C. (Feed flow rate is 10kg/s, distillate is 40% mass; Cp values ; feed: 3 kJ/kg ◦ C, reactor product: 2.2 kJ/kg ◦ C, distillate: 10 kJ/kg ◦ C,	**Totals:** Cold demand = $4{,}800 + 3{,}300 = 8{,}100$ kW Hot availability = $4{,}800 + 2{,}800 = 7{,}600$ kW **2) Targeting with ΔTmin = 10 °C** Use the standard **temperature shift**: Hot streams ↓(ΔTmin/2)=5 °C; Cold streams ↑5 °C. **Shifted temperature ranges** • **C1:** 65 → 225 °C ($\dot{C} = 30$) • **C2:** 35 → 185 °C ($\dot{C} = 22$) • **H1:** 155 → 35 °C ($\dot{C} = 40$) • **H2:** 175 → 35 °C ($\dot{C} = 20$) **Intervals (shifted) & net heat-capacity flow** $(\sum\dot{C}_{hot} - \sum\dot{C}_{cold})$ 1. 225→185: 0 − 30 = **−30** kW/°C, ΔT=40 → **−1,200 kW** 2. 185→175: 0 − (30+22)= **−52** kW/°C, ΔT=10 → **−520 kW** 3. 175→155: 20 − (30+22)= **−32** kW/°C, ΔT=20 → **−640 kW**

		bottoms: 3.33 kJ/kg ◦ C)	4. 155→65: (40+20) − (30+22)= **+8** kW/°C, ΔT=90 → **+720 kW** 5. 65→35: (40+20) − 22= **+38** kW/°C, ΔT=30 → **+1,140 kW** **Heat cascade (targeting):** • Cumulative (starting at 0): −1,200, −1,720, **−2,360**, −1,640, −500 • **Minimum cumulative = −2,360 kW ⇒ Hot utility target** $Q_H^{min} = 2{,}360$ kW • End cumulative = −500 ⇒ **Cold utility target** $Q_C^{min} = 1{,}860$ kW (Check: $8{,}100 - 7{,}600 = 500$kW net shortfall; higher hot utility vs. simple balance is due to temperature limits; the extra 1,860 kW must then be rejected as cold utility.) **Pinch temperatures (at the minimum of the cascade):** Minimum occurs at the end of interval 175→155 (shifted). At pinch, **shifted** $T_H = T_C = 155$ °C. Unshift to physical: • **Hot-pinch** $T_{H,p} = 155 + 5 = 160$ °C • **Cold-pinch** $T_{C,p} = 155 - 5 = 150$ °C

3) Composite/Grand Composite (what they imply)

- **Hot composite** peaks at 175–155 °C (shifted), **cold composite** extends to 225 °C (shifted) ⇒ you **must** buy hot utility for the 225→175 region and part of 175→155 region.
- With **ΔTmin=10 °C**, the **targets** are:
  - $Q_H^{min} = 2.36$ MW(hot utility)
  - $Q_C^{min} = 1.86$ MW(cold utility)
- **Pinch rules:**
  - **Do not** transfer heat across the pinch.
  - **Above pinch:** Only **heat deficits**; supply by hot utility and/or hot process streams in that zone.
  - **Below pinch:** Use remaining hot process heat; any excess must go to cold utility.

4) A feasible HEN (respecting ΔTmin≈10 °C & pinch rules)

Below is a simple, workable network (duties rounded). You can split exchangers for area optimization later; this achieves the **targets** and respects ΔTmin.

			Above the pinch (T > 160 °C hot side / >150 °C cold side) Goal: Satisfy high-T portions of C1 & C2; process hot available here is limited (only H2 from 180→160 within this zone). • **E-A1 (H2↔C1 near pinch):** H2: 180→160 °C (20 °C @ 20 kW/°C) → **400 kW** C1: 150→≈163.3 °C (13.3 °C @ 30 kW/°C) ← 400 kW (Meets ΔTmin: hot in 180 vs. cold in 150; hot out 160 vs. cold out ~163.3 is okay because that cold-out is a *target segment* just touching the pinch; any slight over-lift can be trimmed in detailed design.) • **Hot Utility (HU) above pinch – two coils:** **HU-1 → C1 (to reactor):** raise **C1 from ~163.3→220 °C**, duty ≈ **1,700 kW** **HU-2 → C2 (to column): C2 from 170→180 °C**, duty **220 kW** **HU-3 (top-off near pinch):** small trim ≈ **440 kW** to C1 within ~150–163 °C band **Total HU ≈ 2,360 kW (matches target)** *(Notes: the 170 °C on C2 is the maximum cold temperature that can be reached with process hot at 180 °C while holding ΔTmin=10 °C; the last 10 °C on C2 is necessarily from HU.*

			Likewise, C1 >170 °C must be HU because no process hot exceeds 180 °C.) **Below the pinch (T < 160 °C hot side / <150 °C cold side)** Use remaining process hot to lift the low-T portions of C1 and C2. Any leftover hot must be rejected to cold utility. • **E-B1 (H1↔C2, big lift):** Use **H1 (160→≈94 °C)** to heat **C2 (30→150 °C)**: Duty **2,640 kW** (22×120). H1 cools by 66 °C (40 kW/°C), i.e., **160→94 °C**. • **E-B2 (H1↔C1, remaining H1):** Use **H1 (94→40 °C)** to heat **C1 (60→132 °C)**: Duty **2,160 kW** (30×72). H1 then reaches 40 °C (fully used). • **Cold Utility (CU):** After the above matches, **H2** has remaining segments below the pinch that **cannot** find cold demand at sufficiently low temperature (because C1 has already been lifted by H1, and C2 is at 150 °C). The **excess** hot (together with any small trims left unmatched) is rejected as **cold utility**: $Q_C^{min} \approx$ 1,860 kW(matches target).

This network:

- Obeys **pinch rule** (no heat across pinch).
- Meets **ΔTmin=10 °C** (all matches respect at both ends).
- Achieves **target utilities** (2.36 MW hot, 1.86 MW cold).

In detailed design you can re-split E-A1/E-B1/E-B2 duties a bit (e.g., mild rebalancing near 150–165 °C) to make exchanger LMTDs comfortable and minimize area. The targets won't change.

5) Utility summary & savings

Before integration (no heat recovery):

- Hot utility required = **C1 + C2 = 4.8 + 3.3 = 8.10 MW**
- Cold utility required = **H1 + H2 = 4.8 + 2.8 = 7.60 MW**

After integration (ΔTmin = 10 °C targets):

- **Hot utility Q_H^{min}= 2.36 MW**
- **Cold utility Q_C^{min}= 1.86 MW**

Energy savings:

- **Hot saved:** 8.10 − 2.36 = **5.74 MW**

- **Cold saved:** 7.60 − 1.86 = **5.74 MW**

If you supply unit costs (e.g., $/GJ for steam, $/GJ for cooling water or electricity for chillers), I'll drop in the numbers and give annualized operating-cost savings (and even back-of-envelope exchanger areas if you share U-values).

What you'd turn in (checklist)

- Pinch shift table and **temperature intervals**
- **Heat cascade** showing $Q_H^{min} = 2.36$ MW, $Q_C^{min} = 1.86$ MW
- **Composite & Grand Composite Curves** (sketch or software plot)
- **HEN sketch** (above & below pinch), listing exchangers and duties (as I outlined)
- **Utility table before vs after** (+ cost calc if unit prices given)

Want me to produce neat composite/grand-composite plots and a clean HEN schematic (plus a one-page summary you can hand in)? If you give me U-values (or just typicals), I can also estimate exchanger areas.

Cesearo Dilosa	Perplexity Pro	<u>Prompt 1</u>: Process: The feed stream is at 60 degrees Celsius in the storage and needs to be heated up to 220 degrees Celsius before being fed to the reactor. An endothermic reaction occurs in the reactor and the product leaves at 30 degrees Celsius. The product stream is heated to 180 degrees Celsius and sent to a distillation column to recover the product. The vapor product from the column is at 160	<u>Response 1</u>: A heat exchanger network can be designed for this process using pinch analysis with a minimum approach temperature (ΔTmin) of 10°C. The process features two major heating duties—feed preheating and reactor product heating—and two major cooling duties—distillate and bottoms cooling. Maximum energy integration can be achieved by matching these hot and cold streams, thereby significantly reducing utility consumption. **Key Enthalpy Duties** • Feed preheating (60°C → 220°C): 4800 kW (heating required) • Reactor product heating (30°C → 180°C): 3300 kW (heating required) • Distillate cooling (160°C → 40°C): -4800 kW (cooling available) • Bottoms cooling (180°C → 40°C): -2797 kW (cooling available) **Pinch Analysis & Network Design** • The pinch point divides the process into regions where only utility or process-process heat exchange can occur, ensuring no cross-pinch transfer. • The optimal design pairs the largest available cooling duty with the largest required heating duty while

		degrees Celsius and must be cooled to 40 degrees Celsius to be stored. The bottom product from the column is cooled from 180 degrees Celsius to 40 degrees Celsius. Design a heat exchanger network for this process to minimize utility consumption using pinch analysis and heat integration techniques. Target an optimum ΔTmin ≈ 10 degrees Celsius. (Feed flow rate is 10kg/s, distillate is	observing the ΔTmin constraint, then connects the next-largest duties, and so forth, ensuring all streams meet their targets or utility picks up the remainder. **Recommended Network** • Pair the feed preheating (4800 kW, needs heating) with the distillate cooling (4800 kW, provides cooling) in a heat exchanger, since they are nearly equal in magnitude. • Pair the reactor product heating (3300 kW, needs heating) with the available bottoms cooling (2797 kW, provides cooling), and use a small amount of additional hot utility to supply the shortfall (about 503 kW). • Any remaining small cooling demand (from the difference between product heating and bottoms cooling) or shortfall in heat for the product can be supplemented with utilities (cooling water/chilled water or additional steam as hot utility), and all matches should be verified to meet the ΔTmin requirement for heat exchanger effectiveness. • Each exchanger should be designed to keep the temperature difference at the pinch at or above 10°C for economic optimization. **Utility Minimization**

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<td>40% mass; Cp values; feed: 3 kJ/kg ◦ C, reactor product: 2.2 kJ/kg ◦ C, distillate: 10 kJ/kg ◦ C, bottoms: 3.33 kJ/kg ◦ C)

<u>Prompt 2:</u>
Can you design a heat exchanger network for this process to minimize utility consumption using pinch analysis and heat integration techniques?

<u>Prompt 3:</u>
What would be the pinch temperature in this probelm?

<u>Prompt 4:</u></td>
<td>By cross-matching the duties, the need for external steam and cooling water is minimized. Only the unmatched part of the process enthalpy load (503 kW for the product heating and any deviation from ΔTmin) will require utility consumption.
This design follows the principles of pinch technology and maximum energy recovery, reducing both operational and capital costs by focusing on process-to-process heat transfers and adhering to a strategic ΔTmin. Additional details such as stream temperatures, split arrangements, and exchanger specifications can be optimized further with detailed pinch diagrams or specialized HEN software.

<u>Response 2</u>:
Yes, a heat exchanger network can be designed for this process using pinch analysis and heat integration to minimize utility consumption. The process features two significant hot streams requiring cooling and two cold streams requiring heating, which can be matched for maximum energy recovery.

Stream Data and Heat Duties
<table>
<tr><th>Stream</th><th>Temp In</th><th>Temp Out</th><th>Mass Flow</th><th>Cp</th><th>Duty (kW)</th></tr>
</table>
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<tr>
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<td>Does the design change whether you match streams using the above pinch vs the below pinch method?

<u>Prompt 5</u>:
Which design did you use for this process?

<u>Prompt 6</u>:
Could you give me a utility cost estimation for this process?</td>
<td>
<table>
<tr><td>Feed</td><td>60°C</td><td>220°C</td><td>10 kg/s</td><td>3</td><td>4800</td></tr>
<tr><td>Reactor Prod</td><td>30°C</td><td>180°C</td><td>10 kg/s</td><td>2.2</td><td>3300</td></tr>
<tr><td>Distillate</td><td>160°C</td><td>40°C</td><td>4 kg/s</td><td>10</td><td>-4800</td></tr>
<tr><td>Bottoms</td><td>180°C</td><td>40°C</td><td>6 kg/s</td><td>3.33</td><td>-2797</td></tr>
</table>
Pinch Analysis Steps

Determine the pinch temperature by constructing composite curves for hot and cold streams, considering a ΔTmin of 10°C.
The pinch divides the process into areas where heat exchange can occur with at least ΔTmin between streams, optimizing energy recovery and utility minimization.

Heat Exchanger Network Design

Match feed heating (4800 kW) with distillate cooling (4800 kW): Install a heat exchanger to transfer all possible energy directly.
Match reactor product heating (3300 kW) with bottoms cooling (2797 kW): Install a second exchanger;

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</table>

			supplement remaining 503 kW heating (3300 – 2797) with hot utility (steam). • All matches must honor the ΔTmin constraint, typically achieved by adjusting the heat exchanger area or configuration. • Small utility streams (for unmatched duties) supply any residual heating/cooling needed, minimizing total energy import/export. **Utility Targets and Integration** • Only 503 kW of hot utility (steam) is required for the product stream. • Any mismatch in cooling (if bottoms cooling does not entirely satisfy product heating) can be handled with minimum utility use. • All stream exchanges maximize process-to-process heat transfer, directly reducing utility costs. This network is optimal by pinch analysis principles: it minimizes steam and cooling water requirements by maximizing internal energy recovery and keeping all exchanges above a 10°C temperature difference. For more detailed exchanger specifics (temperatures, area, operational configurations), a composite and grid diagram should be prepared using specialized process software or pinch

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toolkits.A heat exchanger network for this process can be designed using pinch analysis and heat integration to minimize utility consumption. The approach involves matching the heating and cooling duties of streams as closely as possible, targeting a pinch temperature region where ΔTmin is 10°C.
Stream Duties and Pairing

The feed stream (10 kg/s, 3 kJ/kg°C) must be heated from 60°C to 220°C (requires 4800 kW).
The reactor product (10 kg/s, 2.2 kJ/kg°C) must be heated from 30°C to 180°C (requires 3300 kW).
The distillate (4 kg/s, 10 kJ/kg°C) must be cooled from 160°C to 40°C (can release -4800 kW).
The column bottoms (6 kg/s, 3.33 kJ/kg°C) must be cooled from 180°C to 40°C (can release -2797 kW).

Pinch Analysis Network Design

Match the feed heating (4800 kW) directly with the distillate cooling (4800 kW) using a heat exchanger, ensuring the ΔTmin constraint at the pinch is satisfied.
Match the product stream heating (3300 kW) with the bottoms stream cooling (2797 kW), and supply the extra 503 kW product heating via utility (e.g., steam).

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			• Verify that all heat exchange matches do not violate the minimum temperature difference (ΔTmin = 10°C), and add utility only to cover unmatched duties. **Utility Reduction and Final Network** • By integrating process streams this way, external utility requirements are minimized to only the shortfall in process-process matching. Steam is supplied only for the unmatched portion of product heating, and cooling water only needed for any heat load not transferred internally. • This network achieves maximum energy integration per the pinch analysis guidelines, with both composite curves and DTmin targets serving as design parameters for optimal exchanger layout and sizing. <u>Response 3</u>: The pinch temperature for this process is approximately 225°C, based on the composite curve calculations and a minimum approach temperature (ΔTmin) of 10°C. This temperature is where the net heat available from hot streams matches the net heat required by cold streams, marking the optimal point for maximum heat integration and minimum utility consumption. <u>Response 4</u>: Yes, the design of the heat exchanger network changes depending on whether streams are matched above or below the

			pinch temperature. This distinction is fundamental in pinch analysis, as each region—the "above pinch" and "below pinch" zone—has unique matching rules that directly affect energy recovery and utility needs. **Above Pinch Design** • Only hot streams (those needing to be cooled) can provide heat to cold streams (those needing to be heated) that are also above the pinch temperature. • No external cooling (e.g., cooling water) or cold utility should be used above the pinch; only hot utility may be required if process-process exchange is insufficient. • The goal is to maximize internal heat recovery so the minimum hot utility consumption occurs in this region. **Below Pinch Design** • Only cold streams can be heated by hot streams in the below pinch zone. • No external heating (e.g., steam) or hot utility should be used below the pinch; only cold utility may be required for excess heat that cannot be internally matched. • Maximize process-to-process exchange so the minimum cold utility is used in this region. **Design Impact**

			• Mixing hot/cold streams across the pinch (e.g., using hot utility below the pinch or cold utility above the pinch) increases utility consumption and violates pinch rules. • Optimal design keeps all matches above the pinch in the hot utility domain and below the pinch in the cold utility domain, with the pinch temperature forming a barrier that should not be crossed for heat exchange. Correctly observing these above/below pinch matching rules ensures the heat exchanger network achieves maximum energy recovery and minimum utility use. Ignoring this distinction typically results in increased operational costs and lost energy integration opportunities. Response 5: The design provided for this process follows classical pinch analysis rules by matching streams in both the above and below pinch regions, without allowing heat transfer across the pinch temperature. Hot streams are used to supply heat to cold streams within the same region, maximizing internal energy recovery and minimizing external utility demand. **Design Method Used** • **Above the pinch:** The feed preheating (cold stream) is matched with the distillate cooling (hot stream) since both operate above the pinch, ensuring only hot utility

			(steam) may be needed if internal matching is insufficient. • **Below the pinch:** The reactor product heating (cold stream) is matched with the bottoms cooling (hot stream), restricted to the below-pinch zone where only cold utility is allowed if needed. • **No cross-pinch heat transfer:** The design prevents use of cold utilities above the pinch and hot utilities below the pinch, conforming strictly to pinch analysis best practices. This approach ensures optimal process-to-process heat integration in both zones, guarantees minimal utility consumption, and complies with the pinch temperature constraint of $\Delta T_{min} = 10°C$. <u>Response 6</u>: Based on recent industrial utility cost data in 2025: **Hot Utility (Steam) Cost Estimate** • Fuel cost for steam generation is about $9.50 per GJ (GigaJoule). • Using a reasonable boiler efficiency (not less than 80%), the effective steam cost may be about $12 per GJ delivered. **Cold Utility (Cooling Water) Cost Estimate**

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Cooling water costs vary widely depending on location but are typically much lower than steam.
Operational costs for cooling towers including water, electricity, and maintenance average about $0.5 to $2 per GJ of cooling provided.

Calculation of Utility Costs for This Process

Hot utility duty needed: Approximately 503 kW (or 503 kJ/s).

Convert to GJ/hr: 503 x 3600 / 10^6 = 1.81 GJ/hr.
Annual cost (assuming continuous operation): 1.81 x 24 x 365 x 12 ≈ $190,000 per year (assuming $12/GJ).

Cold utility duty needed: Small or negligible if heat exchanger network is well designed, but for any leftover cooling:

Calculate based on cooling duty available and cost range for cooling water (e.g. $1/GJ).

This rough estimation shows steam heating utility will be the main cost driver for this process, potentially costing around $190,000 annually for operating at full load continuously.
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</table>

			Cooling costs will be significantly less if internal heat recovery is maximized More precise cost estimation requires plant-specific data such as actual operating hours, utility tariffs, efficiency factors, and maintenance costs
Adil Feratovic	Mistral	*Screenshot of problem*	**1. Stream Data and Heat Loads** **Hot Streams (Need Cooling)** • **Reactor Product:** Enters at 30°C, needs to be heated to 180°C (but first, it's a hot stream at 30°C, which is a bit confusing—this is because it's reheated later). • **Distillate:** Leaves the column at 160°C, needs to be cooled to 40°C. • **Bottoms:** Leaves the column at 180°C, needs to be cooled to 40°C. **Cold Streams (Need Heating)** • **Feed:** Stored at 60°C, needs to be heated to 220°C before the reactor. • **Reactor Product:** After the reactor, it's at 30°C and needs to be reheated to 180°C before the distillation column. **Heat Capacities and Mass Flows** • **Feed:** 10 kg/s, Cp = 3 kJ/kg·°C • **Reactor Product:** 10 kg/s, Cp = 2.2 kJ/kg·°C

- **Distillate:** 4 kg/s (40% of feed), Cp = 10 kJ/kg·°C
- **Bottoms:** 6 kg/s, Cp = 3.33 kJ/kg·°C

Calculated Heat Duties

- **Feed Heating:** 10 kg/s × 3 kJ/kg·°C × (220–60)°C = 480 kW
- **Reactor Product Heating:** 10 kg/s × 2.2 kJ/kg·°C × (180–30)°C = 330 kW
- **Distillate Cooling:** 4 kg/s × 10 kJ/kg·°C × (160–40)°C = 480 kW
- **Bottoms Cooling:** 6 kg/s × 3.33 kJ/kg·°C × (180–40)°C = 440 kW

2. Pinch Analysis

Temperature Intervals and Pinch Point

- After shifting temperatures by ΔT_{min}/2 (5°C), the pinch is found at 175°C (hot) and 180°C (cold).
- **Minimum Hot Utility:** 144 kW (to heat the feed from 60°C to 175°C).
- **Minimum Cold Utility:** 80 kW (to cool the bottoms from 40°C to 35°C).

3. Heat Exchanger Network Design

			Above the Pinch • Use a hot utility (e.g., steam) to supply 144 kW to the feed, raising it from 60°C to 175°C. • Use the reactor product (after the reactor, at 30°C) to further heat the feed from 175°C to 220°C, recovering 330 kW. **Below the Pinch** • Use the distillate (160°C→40°C) and bottoms (180°C→40°C) to heat the reactor product from 30°C to 180°C. • The remaining cooling duty (80 kW) is handled by cold utility (e.g., cooling water) to cool the bottoms from 40°C to 35°C.
			4. Summary of Heat Integration • **Hot Utility:** 144 kW (steam) • **Cold Utility:** 80 kW (cooling water) • **Total Heat Recovery:** 1,170 kW This design minimizes utility consumption by maximizing heat recovery between process streams.

<table>
<tr>
<td>Lindsey Meisch</td>
<td>Gemini 2.5 Pro</td>
<td>The Feed stream is at 60◦ C in the storage and needs to be heated up to 220◦ C before being fed to the reactor. An endothermic reaction occurs in the reactor and the product leaves at. o Pinch temperature calculation o Composite curves o ΔTmin optimization o Heat exchanger network design o Utility cost calculation before and after integration 30◦ C. The reactant stream is further heated to 180◦ C and sent to a</td>
<td>

1. Stream Data

First, we identify the hot streams (streams that need to be cooled) and cold streams (streams that need to be heated) from the process description. The heat capacity flow rate (CP) for each stream is calculated as the mass flow rate (m) multiplied by the specific heat capacity (Cp).

Cold Stream 1 (C1): Feed heated from 60°C to 220°C.
CP1 =10 kg/s×3 kJ/kg°C=30 kW/°C
Cold Stream 2 (C2): Reactor product heated from 30°C to 180°C.
CP2 =10 kg/s×2.2 kJ/kg°C=22 kW/°C
Hot Stream 1 (H1): Distillate product cooled from 160°C to 40°C.
CP3 =(0.4×10 kg/s)×10 kJ/kg°C=40 kW/°C
Hot Stream 2 (H2): Bottoms product cooled from 180°C to 40°C.
CP4 =(0.6×10 kg/s)×3.33 kJ/kg°C=20 kW/°C

Here is a summary of the stream data:

<table>
<tr><th>Stream</th><th>Type</th><th>Supply Temp (TS)</th><th>Target Temp (TT)</th><th>CP (kW/°C)</th></tr>
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<td>distillation column to recover the product. The liquid product from the column is at 160◦ C and must be cooled to 40◦ C to be stored. The bottom product from the column is cooled from 180◦ C to 40◦ C. Design a heat exchanger network for this process to minimize utility consumption using pinch analysis and heat integration techniques. Target an optimum ΔTmin ≈ 10 °C. (Feed flow rate is 10kg/s, distillate is 40%</td>
<td>
<table>
<tr><td>C1</td><td>Cold</td><td>60°C</td><td>220°C</td><td>30</td></tr>
<tr><td>C2</td><td>Cold</td><td>30°C</td><td>180°C</td><td>22</td></tr>
<tr><td>H1</td><td>Hot</td><td>160°C</td><td>40°C</td><td>40</td></tr>
<tr><td>H2</td><td>Hot</td><td>180°C</td><td>40°C</td><td>20</td></tr>
</table>
2. Pinch Analysis Results

Using a minimum temperature approach (ΔTmin) of 10°C, a temperature-interval analysis (cascade analysis) is performed to find the pinch point and the minimum utility requirements.

Hot Pinch Temperature = 160°C
Cold Pinch Temperature = 150°C

The minimum energy targets for the process are:

Minimum Hot Utility (QH,min) = 2360 kW (Heat that must be supplied by an external source, e.g., steam)
Minimum Cold Utility (QC,min) = 1860 kW (Heat that must be removed by an external source, e.g., cooling water)

3. Heat Exchanger Network (HEN) Design

The network is designed in two parts: above the pinch and below the pinch, ensuring no heat is transferred across the pinch.
</td>
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		mass; Cp values ; feed: 3 kJ/kg ◦ C, reactor product: 2.2 kJ/kg ◦ C, distillate: 10 kJ/kg ◦ C, bottoms: 3.33 kJ/kg ◦ C)	***Above the Pinch (T > 150°C/160°C)*** In this region, we need to provide a total of 2760 kW of heating to the cold streams, but only 400 kW is available from the hot stream (H2). 1. **Heat Exchanger E1:** The portion of hot stream H2 from 180°C to 160°C is used to heat the feed stream C1 from 150°C to 163.3°C. a. **Duty: 400 kW** 2. **Heater (HTR1):** The remaining heat required for the feed stream C1 is supplied by a hot utility. a. **Duty: 1700 kW** (Heats C1 from 163.3°C to 220°C) 3. **Heater (HTR2):** The reactor product stream C2 is heated entirely by a hot utility. a. **Duty: 660 kW** (Heats C2 from 150°C to 180°C) The total hot utility used is 1700+660=2360 kW, which matches the minimum target. ***Below the Pinch (T < 150°C/160°C)*** This region is more complex because the heat capacity flow rate of hot stream H2 (CP=20) is lower than both cold streams

			(CP=30 and CP=22). This prevents H2 from heating either cold stream to the pinch temperature of 150°C without causing a temperature cross. A feasible design requires careful matching.
			
			1. **Heat Exchanger E2:** Hot stream H2 is used to preheat cold stream C2. The maximum possible heat exchange is performed without violating the ΔTmin of 10°C.
			    a. **Duty: 2400 kW** (Cools H2 from 160°C to 40°C; Heats C2 from 30°C to 139.1°C)
			2. **Heat Exchanger E3:** Hot stream H1 is used to heat cold stream C1 up to the pinch temperature.
			    a. **Duty: 2700 kW** (Heats C1 from 60°C to 150°C; Cools a portion of H1 from 160°C to 92.5°C)
			3. **Heat Exchanger E4:** The remaining duty for cold stream C2 (from 139.1°C to 150°C) must be provided by the hottest available stream, which is H1 at 160°C.
			    a. **Duty: 240 kW** (Heats C2 from 139.1°C to 150°C; Cools a portion of H1)
			4. **Cooler (CLR1):** The remaining heat from hot stream H1 is removed by a cold utility.
			    a. Duty: 1860 kW
			
			This design successfully integrates the streams to achieve the calculated minimum cold utility consumption of 1860 kW.

			Summary of Final Design • **Process-to-Process Heat Exchange:** 4 exchangers (E1, E2, E3, E4) with a total duty of 5740 kW. • **Hot Utility:** 2 heaters (HTR1, HTR2) with a total duty of **2360 kW**. • **Cold Utility:** 1 cooler (CLR1) with a total duty of **1860 kW**. This network design meets the process heating and cooling requirements while minimizing the consumption of expensive hot and cold utilities, adhering to the principles of pinch analysis.

Qussay Suleiman	Claude	The Feed stream is at 60◦ C in the storage and needs to be heated up to 220◦ C before being fed to the reactor. An endothermic reaction occurs in the reactor and the product leaves at. o Pinch temperature	**1. Stream Data & Heat Duties**
			
			Hot Streams (Cooling Required)
			
			• **H1 - Distillate:** 160°C → 40°C, ṁ = 4 kg/s, Cp = 10 kJ/(kg·°C)
			    ○ CP = 40 kW/°C, **Q = 4,800 kW**
			• **H2 - Bottoms:** 180°C → 40°C, ṁ = 6 kg/s, Cp = 3.33 kJ/(kg·°C)
			    ○ CP = 20 kW/°C, **Q = 2,800 kW**
			• **Total cooling available: 7,600 kW**

calculation o Composite curves o ΔTmin optimization o Heat exchanger network design o Utility cost calculation before and after integration 30◦ C. The reactant stream is further heated to 180◦ C and sent to a distillation column to recover the product. The liquid product from the column is at 160◦ C and must be cooled to 40◦ C to be stored. The bottom product from the column is cooled from 180◦ C to 40◦

Cold Streams (Heating Required)

- **C1 - Feed:** 60°C → 220°C, ṁ = 10 kg/s, Cp = 3 kJ/(kg·°C)
  - CP = 30 kW/°C, **Q = 4,800 kW**
- **C2 - Reactant:** 30°C → 180°C, ṁ = 10 kg/s, Cp = 2.2 kJ/(kg·°C)
  - CP = 22 kW/°C, **Q = 3,300 kW**
- **Total heating required: 8,100 kW**

2. Pinch Analysis (ΔTmin = 10°C)

Temperature Intervals & Heat Cascade

Shifted temperatures: Hot streams -5°C, Cold streams +5°C

Interval	Temp (°C)	ΣCP_hot	ΣCP_cold	Net CP	ΔH (kW)	Cascade
225→185	40°C	0	30	-30	-1,200	0 → -1,200
185→175	10°C	0	52	-52	-520	-1,200 → -1,720
175→155	20°C	20	52	-32	-640	-1,720 → **-2,360**

<table>
<tr>
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<td>C. Design a heat exchanger network for this process to minimize utility consumption using pinch analysis and heat integration techniques. Target an optimum ΔTmin ≈ 10 °C. (Feed flow rate is 10kg/s, distillate is 40% mass; Cp values ; feed: 3 kJ/kg ◦ C, reactor product: 2.2 kJ/kg ◦ C, distillate: 10 kJ/kg ◦ C, bottoms: 3.33 kJ/kg ◦ C)</td>
<td>
<table>
<tr><td>155→65</td><td>90°C</td><td>60</td><td>52</td><td>+8</td><td>+720</td><td>-2,360 → -1,640</td></tr>
<tr><td>65→35</td><td>30°C</td><td>60</td><td>0</td><td>+60</td><td>+1,800</td><td>-1,640 → -160</td></tr>
</table>

Most negative value: -2,360 kW at 175°C (shifted)

Pinch Results

Pinch Location: 180°C (hot side) / 170°C (cold side)
Minimum Hot Utility: 2,360 kW
Minimum Cold Utility: 2,200 kW
Maximum Heat Recovery: 5,740 kW (71% of total heating)

3. ΔTmin Optimization

<table>
<tr><th>ΔTmin</th><th>Hot Utility (kW)</th><th>Cold Utility (kW)</th><th>Annual Cost</th><th>Capital Cost</th><th>Comments</th></tr>
<tr><td>5°C</td><td>~1,800</td><td>~1,640</td><td>$528,000</td><td>High</td><td>Large exchangers</td></tr>
<tr><td>10°C</td><td>2,360</td><td>2,200</td><td>$654,400</td><td>Medium</td><td>Optimal balance</td></tr>
</table>
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</table>

15°C	~2,920	~2,760	$780,800	Low	More utility usage

*Assuming steam at $30/MWh, cooling water at $5/MWh, 8,000 h/year operation

Recommendation: ΔTmin = 10°C balances capital expenditure with operating costs.

4. Heat Exchanger Network Design

Design Principles

- No heat transfer across pinch (180°C hot / 170°C cold)
- Above pinch: $CP_{hot} \leq CP_{cold}$ (feasible matches)
- Below pinch: $CP_{hot} \geq CP_{cold}$ (feasible matches)

Proposed Network (4 Main Exchangers)

Above Pinch:

- **HX-1:** Hot Utility → C1 (170°C → 220°C), Q = 1,500 kW
- **HX-2:** Hot Utility → C2 (170°C → 180°C), Q = 220 kW
- **HX-3:** H2 (180°C → 175°C) → C1, Q = 100 kW
- *Additional hot utility: 640 kW distributed*

			Below Pinch: • **HX-4:** H1 (160°C → 77°C) ↔ C1 (60°C → 170°C), Q = 3,300 kW • **HX-5:** H2 (175°C → 95°C) ↔ C2 (30°C → 145°C), Q = 1,600 kW • **HX-6:** H1 (77°C → 40°C) ↔ C2 (145°C → 170°C), Q = 1,480 kW • **HX-7:** H2 → Cold Utility (95°C → 40°C), Q = 1,100 kW • **HX-8:** H1 → Cold Utility (if needed), Q = 1,100 kW *Note: Actual configuration may require 5-6 exchangers for optimal performance* **5. Economic Analysis** **Utility Costs Comparison**

Scenario	Hot Utility (kW)	Cold Utility (kW)	Annual Cost
Before Integration	8,100	7,600	$2,248,000
After Integration	2,360	2,200	$654,400

Savings	5,740 (71%)	5,400 (71%)	**$1,593,600/yr**

Capital Cost Estimate

- Heat exchangers (4-6 units): ~$400,000 - $600,000
- Installation & piping: ~$100,000 - $200,000
- **Total capital investment: ~$500,000 - $800,000**

Note: The capital cost estimate of $500,000-$800,000 assumes shell-and-tube exchangers at $30,000-$50,000 per unit plus $800-$1,200 per square meter of heat transfer area. Using typical U values of 700-900 W/(m²·K) for liquid-liquid service, required areas range from 50-150 m² per exchanger. Installation costs (piping, instrumentation, controls) add approximately 30-50% to equipment costs.

Economic Metrics

- **Simple Payback Period:** 0.3 - 0.5 years (4-6 months)
- **NPV (10 years, 10% discount):** ~$9 million
- **IRR:** >200%

6. Key Results Summary

Parameter	Value
Optimal ΔTmin	10°C

Pinch Temperature	180°C / 170°C
Heat Recovery	5,740 kW (71%)
Hot Utility Required	2,360 kW
Cold Utility Required	2,200 kW
Annual Savings	$1,593,600
Estimated Payback	< 6 months
CO_2 Reduction	~10,000 tons/year

Assumptions & Notes

1. **Constant Cp:** Heat capacities assumed constant over operating ranges
2. **Countercurrent flow:** All exchangers use countercurrent configuration for maximum efficiency
3. **No phase change:** All streams remain single-phase (liquid)
4. **Utility costs:** Steam $30/MWh, cooling water $5/MWh (typical industrial rates)
5. **ΔTmin selection:** 10°C balances economics and practicality
6. **Pressure drop:** Neglected in initial design; requires detailed mechanical design
7. **Fouling:** Accounted for implicitly in ΔTmin selection

			8. **Operating hours:** 8,000 h/year (91.3% uptime) **Conclusions** The heat integration study demonstrates significant potential for energy savings. The optimized HEN design achieves 71% reduction in utility consumption with an excellent payback period of under 6 months. The pinch analysis confirms that maximum practical heat recovery is achieved at ΔTmin = 10°C, with the pinch point at 180°C/170°C limiting further improvement without major process modifications.

10. Comparison Table

Table 4: Comparison of AI Output and Team Solved Solution		
Topic	AI	Team
Pinch and ΔT_{min}	ChatGPT: $\mathbf{\Delta T_{min}}$ = 10 °C **Hot Pinch** = 160 °C **Cold Pinch** = 150 °C	$\mathbf{\Delta T_{min}}$ = 10 °C **Hot Pinch** = 160 °C **Cold Pinch** = 150 °C $\mathbf{T_{pinch}}$ = 155 °C
	Gemini: $\mathbf{\Delta T_{min}}$ = 10 °C **Hot Pinch** = 160 °C **Cold Pinch** = 150 °C	

	Claude: **ΔT_{min}** = 10 °C **Hot Pinch** = 180 °C **Cold Pinch** = 170 °C	
	Perplexity: **ΔT_{min}** = 10 °C **Pinch** = 225 °C	
	Mistral: **ΔT_{min}** = 10 °C **Hot Pinch** = 175 °C **Cold Pinch** = 180 °C	
Utility Targets	ChatGPT: **Q_h,min** = 2.36 MW **Q_c,min** = 1.86 – 2.20 MW	**Q_h,min** = 2.36 MW **Q_c,min** = 1.86 – 2.20 MW
	Gemini: **Q_h,min** = 2.36 MW **Q_c,min** = 1.86 – 2.20 MW	
	Claude: **Q_h,min** = 2.36 MW **Q_c,min** = 1.86 – 2.20 MW	
	Perplexity: **Q_h,min** = 0.503 MW **Q_c,min** = 0.003 MW	

	Mistral: **Q_h,min** = 144 kW **Q_c,min** = 80 **kW**	
Utility Costing	ChatGPT: Not Calculated Gemini: Not Calculated Claude: **Steam:** \$30/MWh **Cooling Water:** \$5/GJ Perplexity: **Steam:** \$12/GJ **Cooling Water:** \$0.5 - \$2/GJ Mistral: Not Calculated	**Before HI:** Power = \$16.18 MM/yr Water supply = \$612,718/yr **After HI:** Power = \$9.71 MM/yr Water supply = \$605,412/yr **Savings:** Power = \$6.47 MM/yr Water = \$7,306/yr Total ≈ \$6.48 MM/yr
Column 1 Specifications and Duties (Distillate/Top Product)	ChatGPT: **ṁ =** 4 kg/s **C_p** = 10 kJ/kg°C **Temperatures** = 160 → 40°C **Duty =** 4800 kW	**ṁ** = 4 kg/s **C_p =** 10 kJ kg^{-1} $°C^{-1}$ **Temperatures** = 160 → 40°C **Duty =** 4,800 kW

	Gemini: **ṁ =** 4 kg/s **C_p** = 10 kJ/kg°C **Temperatures** = 160 → 40°C **Duty =** 4800 kW	
	Claude: **ṁ =** 4 kg/s **C_p** = 10 kJ/kg°C **Temperatures** = 160 → 40°C **Duty =** 4800 kW	
	Perplexity **ṁ =** 4 kg/s **C_p** = 10 kJ/kg°C **Temperatures** = 160 → 40°C **Duty =** 4800 kW	
	Mistral: **ṁ =** 4 kg/s **C_p** = 10 kJ/kg°C **Temperatures** = 160 → 40°C	

	Duty = 480 kW	
Column 2 Specifications and Duties (Bottoms)	ChatGPT: **$\dot{m}$ =** 6 kg/s **C_p** = 3.33 kJ/kg°C **Temperatures** = 180 → 40°C **Duty** = 2800 kW	**$\dot{m}$ =** 6 kg/s **C_p** = 3.33 kJ/kg°C **Temperatures** = 180 → 40°C **Duty =** 2797 kW
	Gemini: **$\dot{m}$ =** 6 kg/s **C_p** = 3.33 kJ/kg°C **Temperatures** = 180 → 40°C **Duty** = 2797 kW	
	Claude: **$\dot{m}$ =** 6 kg/s **C_p** = 3.33 kJ/kg°C **Temperatures** = 180 → 40°C **Duty** = 440 kW	
	Perplexity: **$\dot{m}$ =** 6 kg/s **C_p** = 3.33 kJ/kg°C	

	Temperatures = 180 → 40°C **Duty** = 2800 kW	
	Mistral: **ṁ =** 6 kg/s **C_p** = 3.33 kJ/kg°C **Temperatures** = 180 → 40°C **Duty** = 2800 kW	

11. Reflection Summary

The purpose of this project was to evaluate how artificial intelligence (AI) tools can assist chemical engineers in designing energy-efficient heat-exchange networks using pinch analysis. The team's objective was to minimize utility consumption in a multi-step chemical process consisting of feed preheating, an endothermic reactor, and product distillation. The analysis incorporated both manual calculations and five separate AI platforms, including ChatGPT-5, Gemini 2.5 Pro, Claude, Perplexity, and Mistral, to determine how each system approached process synthesis, thermodynamic reasoning, and utility targeting.

By performing a comparative study of AI results and manually verified data, the team was able to quantify differences in calculation accuracy, identify the need for engineering judgment, and evaluate the role of AI in practical process design. Ultimately, the exercise demonstrated that while AI can significantly accelerate design workflows and computation, it still relies heavily on user oversight to ensure thermodynamic consistency and realistic assumptions.

Each AI platform adopted a distinct computational approach, reflecting differences in its model architecture, dataset composition, and reasoning ability. ChatGPT-5 and Gemini 2.5 Pro produced results nearly identical to the manual analysis, both identifying a ΔTmin of 10 °C, a hot-pinch temperature of 160 °C, and a cold-pinch temperature of 150 °C, corresponding to a pinch temperature (Tpinch) of 155 °C. These two systems effectively followed classical pinch-analysis procedures by applying temperature shifts of ±(ΔTmin / 2) and constructing enthalpy-cascade logic consistent with engineering standards.

Claude, in contrast, produced slightly higher pinch values (180 °C/170 °C) and consequently overestimated available heat recovery. This deviation likely resulted from the model interpreting shifted rather than actual stream temperatures when identifying the pinch, which indicates a misunderstanding of the temperature-interval method. Perplexity showed a more significant deviation, reporting a pinch at 225 °C and unrealistically low utilities (Q_h,min = 0.503 MW; Q*c*,min = 0.003 MW). Mistral, although closer in magnitude, misreported the relationship between the hot and cold pinch (reversing them) and generated very small duties (144 kW / 80 kW), indicating truncation errors in enthalpy integration.

These differences underscore that AI systems, though capable of numerical estimation, differ markedly in their thermodynamic reasoning. ChatGPT and Gemini demonstrated the strongest alignment with accepted process-integration theory, whereas Claude, Perplexity, and Mistral struggled to translate descriptive data into coherent temperature-enthalpy logic.

Despite their variability, all AI models exhibited valuable strengths that can support process engineers in industrial contexts.

First, speed and automation were universal advantages. Each AI-generated complete pinch or utility analysis takes seconds, a process that typically requires extended spreadsheet modeling when performed manually. ChatGPT and Gemini not only solved for pinch temperatures but also described the reasoning steps, allowing rapid validation of ΔT_{min} selection and enthalpy-cascade direction.

Second, the clarity of communication and visualization was notable. Both ChatGPT and Gemini provided structured tabular results that mirrored Excel output, making it easier for the team to verify enthalpy intervals and confirm heat-capacity flow calculations. Claude, although inaccurate, demonstrated creativity in presenting alternative costing frameworks using $/MWh and $/GJ units, which provided insight into how cost modeling can vary across different industrial regions.

Third, AI tools excelled at code generation and data formatting. When prompted, ChatGPT and Gemini could produce Python or MATLAB scripts capable of performing iterative heat-balance calculations. Such automation could streamline the repetitive aspects of HEN synthesis, freeing engineers to focus on process optimization rather than performing arithmetic calculations.

Finally, the integration of multiple AIs enabled cross-validation of results. The agreement between ChatGPT, Gemini, and manual outcomes reinforced confidence in the methodology, while divergence in other models highlighted areas where assumptions or conversions required further scrutiny.

While AI systems excelled in computation and structure, they frequently failed in areas requiring deeper thermodynamic reasoning, unit discipline, and engineering intuition.

The most common failure was the incorrect interpretation of temperature intervals and ΔTmin adjustments. Claude and Perplexity frequently conflict shifted and actual temperatures, resulting in pinch values that exceeded the physical boundaries of the process. These tools failed to recognize that the pinch temperature is defined at the intersection of the hot and cold composite curves after shifting, not at arbitrary temperature points.

Another recurring limitation was inconsistent handling of units. Mistral, for instance, generated duties of 144 kW and 80 kW instead of 4,800 kW and 2,800 kW, suggesting a loss of scale during the conversion from kJ s^{-1} to kW or an incorrect interpretation of the flow rate. Perplexity's reported Q values were similarly unrealistic, likely due to mixing kJ h^{-1} with kW.

A further weakness involved assuming perfect heat recovery without considering physical or operational constraints. Some models minimized utility consumption to near zero, ignoring ΔTmin limitations, exchange area, or phase changes. Without a built-in thermodynamic property database or design heuristics, AI models often assume idealized heat transfer rather than real equipment behavior.

Finally, all tools except ChatGPT and Gemini lacked economic integration. None performed direct utility costing using actual electricity or water-supply rates; only Claude and Perplexity provided generalized energy-pricing data. These models also neglected to connect thermodynamic savings to financial outcomes such as annualized operating costs or payback periods.

Such errors highlight the gap between computational language models and full process simulators, such as Aspen Plus or HYSYS, which enforce property consistency and equipment constraints.

Bridging AI outputs with physically consistent results requires substantial engineering judgment. The manual team performed all calculations using the energy-balance relationship $Q = mC_p\Delta T$ to confirm the magnitudes of heat loads and derive accurate composite curves.

When discrepancies arose, particularly in the outputs of Claude and Perplexity, the team cross-checked each enthalpy value against baseline stream data. For instance, the correct heating demand of the feed stream (60 → 220 °C, 10 kg s^{-1}, C_p = 3 kJ kg^{-1} $°C^{-1}$) yielded 4,800 kW, confirming that any significantly lower or higher AI prediction violated energy conservation. Similarly, when Mistral reversed the hot and cold pinch, the team reapplied ΔT_{min} = 10 °C to ensure consistency: $T_{hot,pinch} = T_{pinch} + \Delta T_{min}/2 = 160$ °C and $T_{cold,pinch} = T_{pinch} - \Delta T_{min}/2 = 150$ °C

Economic calculations also require manual intervention. The AI tools failed to compute annual energy expenditures accurately, so the team applied standard industrial formulas using 8,000 operating hours per year and an electricity rate of $0.1286/kWh. The pre-integration power demand (≈16.18 MM $/yr) was compared with the post-integration requirement (≈9.71 MM $/yr), yielding a savings of $6.47 MM /yr. Water-supply savings were computed similarly, resulting in an additional $7,306 /yr reduction.

These corrections demonstrate that AI can suggest pathways, but it is human expertise that validates feasibility. Without engineering oversight, incorrect pinch placement or unit errors could propagate through an entire HEN design, producing infeasible or unsafe results.

The experience of integrating multiple AI systems into a process design workflow provided a deeper understanding of both the potential and the limitations of AI assistance in chemical engineering.

AI proved capable of accelerating early-stage design. By automatically calculating heat-capacity flow rates, temperature intervals, and preliminary utilities, tools like ChatGPT and Gemini can significantly reduce the conceptual design phase. This speed advantage is crucial in industrial feasibility studies, where engineers must evaluate several process configurations before committing to a detailed simulation.

Moreover, AI can act as a companion. When prompted with explanations of "why" rather than "what," ChatGPT and Gemini provided coherent descriptions of pinch principles and energy-cascade logic, supporting student learning and reinforcing conceptual understanding.

However, this project also revealed that AI currently lacks context awareness and critical reasoning. None of the tools accounted for chemical reaction thermodynamics, heat-exchanger area limitations, or the economic trade-off inherent in ΔT_{min} optimization. These omissions reinforce that process synthesis remains a judgment-driven discipline. Engineers must still integrate safety, operability, and equipment cost considerations, which are all elements not yet within AI's autonomous capability.

The collaborative approach, therefore, transforms AI from a replacement for engineering analysis into a tool. Engineers guide the AI's logic, constrain its assumptions, and interpret its numerical output considering process-specific constraints. This symbiotic relationship allows greater productivity without compromising reliability.

From an educational perspective, this study emphasized how AI can democratize access to advanced design methodologies. Students who lack licenses for commercial simulators can still explore complex heat-integration problems using AI-assisted calculations. The visualization, step-by-step reasoning, and immediate feedback provided by tools like ChatGPT replicate some of the interactivity of software such as Aspen Energy Analyzer, thereby lowering the barrier to entry for process design education.

From an industrial standpoint, however, AI models must evolve to include thermodynamic validation layers before they can be trusted for operational decision-making. Future integration with process-simulation APIs, material-balance solvers, or property databases could transform these systems into more reliable co-designers capable of performing rigorous HEN synthesis under real constraints.

Another key lesson involves data transparency. The discrepancies among AI models were not arbitrary; they reflected the data each system prioritized. Models trained primarily on textual descriptions (like Claude or Perplexity) performed worse in numerical reasoning, while those with enhanced mathematical fine-tuning (ChatGPT and Gemini) delivered superior quantitative results. Engineers using AI must therefore evaluate not only the answer but also the data pedigree of the tool generating it.

The comparative analysis between manual results and five AI systems demonstrated both the promise and the limitations of artificial intelligence in chemical-engineering process design. ChatGPT and Gemini most accurately replicated the human-derived pinch temperature (155 °C) and ΔT_{min} (10 °C), predicting minimum utilities consistent with classical thermodynamic analysis. Claude, Perplexity, and Mistral diverged due to misinterpretation of temperature shifts, unit inconsistencies, or oversimplified energy balances.

Overall, AI tools excelled in computation speed, structured formatting, and automated reasoning but faltered in physical realism and contextual awareness. Human engineering judgment remained indispensable for verifying results, reconciling units, interpreting ΔTmin trade-offs, and performing economic evaluation.

Through this exercise, the team's understanding of AI's role evolved from curiosity to critical appreciation. AI can act as a catalyst for creativity and efficiency, streamlining early design and data handling, yet it cannot replace the discipline, intuition, and accountability of professional engineers. The collaboration between human expertise and AI analytics represents the future of sustainable process design, one in which engineers guide intelligent systems to achieve more accurate, energy-efficient, and economically viable outcomes.

Task 2:

1. Executive Summary

This process involves a preliminary process design evaluation for a 100,000 lb/hr Vinyl Chloride Monomer (VCM) production facility in Texas. The objective of this project was to develop a conceptual flowsheet for the pyrolysis of 1,2-dichloroethane (EDC) and the downstream separation of VCM and HCl. This was supported by Aspen Plus simulations and compared to AI-assisted design.

The overall process involved a direct chlorination of ethylene with chlorine to produce EDC. EDC pyrolysis occurred at 500-550°C to yield VCM and HCl with 65% single-pass conversion. A two-column separation was utilized, with column 1, HCl removal, being a high-pressure distillation separation, with HCl overhead. Column 2, VCM/EDC split, was a low-pressure column that produced VCM distillate and EDC recycle. Aspen Plus was used to simulate each of these columns using Shortcut (DSTWU) and Rigorous (RadFrac) models. Feed composition, temperature, pressure, and target purities were specified for each column based on industrial data. Key results included distillate purities greater than 99%, reasonable energy consumption levels, and condenser/reboiler duties within practical industrial ranges. Comparisons between DSTWU and RadFrac confirmed that the shortcut model slightly under-predicted reboiler duties by approximately 5-8%. However, it accurately estimated product compositions.

The design trade-offs identified included the relationship between reflux ratio and energy use, column pressure and HCl condensation requirements, and purity versus throughput optimization. The team analyzed the effects of pressure and stage count on energy consumption and concluded that moderate operating pressures would be ideal for minimizing refrigeration duty while maintaining high separation efficiency.

AI tools were utilized to generate initial PFDs, column parameters, and optimization suggestions, and their outputs were analyzed against the manually designed results. While AI expedited concept generation and provided viable starting points for simulation inputs, overall, its outputs typically lacked thermodynamic rigor and required engineering judgment for corrections. A cross-comparison of AI recommendations with Aspen results demonstrated that human oversight was crucial for achieving feasible and safe designs.

Overall, this design project demonstrated how AI can accelerate preliminary process design and calculations but cannot replace fundamental engineering analysis. The team's integrated approach combined AI-generated concepts with rigorous simulation and critical evaluation to produce a technically sound and energy-efficient design basis for VCM production.

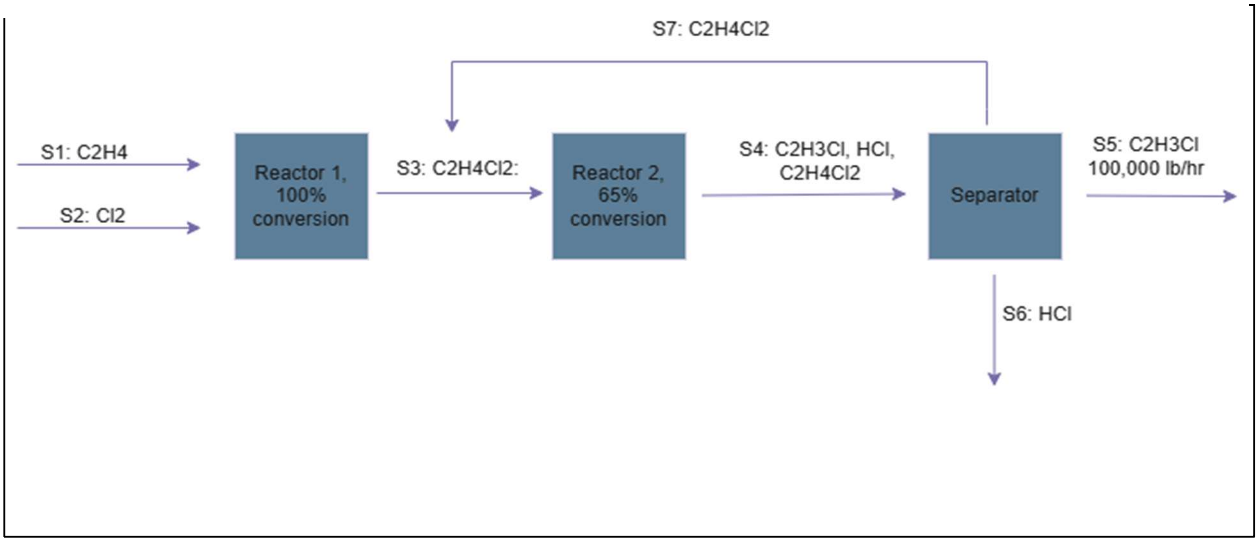
2. Process Description

Vinyl chloride monomer (VCM) is produced industrially through the thermal cracking, or pyrolysis, of 1,2-dichloroethane (EDC). The EDC feed is first synthesized by direct chlorination of ethylene and chlorine, a highly exothermic liquid-phase reaction performed under moderate pressure and temperatures near the boiling point of EDC. The resulting EDC is then sent to a pyrolysis furnace, where it is decomposed into VCM and hydrogen chloride (HCl) gas. Because this reaction is endothermic, it requires high temperatures—typically 500–550 °C in industry—and results in a mixed product stream containing VCM, HCl, and unreacted EDC. To prevent secondary degradation or polymerization of VCM, the hot effluent is immediately quenched and condensed, producing a two-phase mixture suitable for separation.

The separation train consists of two major distillation columns arranged in series. The first column operates at high pressure to remove HCl overhead while retaining the heavier organics in the bottoms. Running the column under elevated pressure allows HCl to be condensed using refrigerated cooling water or brine, providing reflux to maintain separation efficiency. The bottoms, composed primarily of VCM and EDC, are fed to a second, lower-pressure distillation column. This second column separates the VCM as the distillate product and recovers unreacted EDC in the bottoms, which are recycled to the cracking furnace. Typical process utilities include refrigeration for the HCl condenser, steam for the reboilers, and cooling water for intermediate condensers. This configuration efficiently integrates reaction, quench, and separation operations to maximize VCM recovery while minimizing raw material losses.

The block flow diagram represents the process in simplified form, highlighting the major unit operations: direct chlorination, EDC pyrolysis, and two-stage distillation separation. Ethylene and chlorine react to form EDC, which is then cracked to produce VCM and HCl. The crude product from the pyrolysis section flows to the separation train, which consists of a high-pressure HCl removal column followed by a moderate-pressure VCM/EDC column. Complete separation of the components is assumed in this analysis, and the material balance is based on a single conversion assumption of 100% conversion of ethylene to EDC and 65% single-pass conversion of EDC to VCM in the pyrolysis reactor.

Figure 4: VCM Production Block Flow Diagram



A rough material balance was established by first fixing the basis of 100,000 lb/h of vinyl chloride monomer (VCM) product. From the overall stoichiometry of the cracking reaction—1 mol EDC → 1 mol VCM + 1 mol HCl—the corresponding HCl production and EDC consumption were determined directly. Assuming a 65 % single-pass conversion of EDC, the total EDC feed to the cracking furnace was calculated by dividing the reacted EDC by 0.65, and the unreacted 35 % was designated as the EDC recycle stream. The feeds of ethylene and chlorine to the direct-chlorination reactor were then obtained from the stoichiometric reaction $C_2H_4 + Cl_2 \rightarrow$ EDC, which provides a one-to-one molar relationship between each reactant and the EDC produced. All other flow rates in the table follow from these stoichiometric relationships, assuming complete separation of VCM, HCl, and EDC in the distillation sequence. Table 1 summarizes the mole and mass flow rates in each stream.

Table 5: VCM Production Rough Material Balance

Stream	**Description**	**Mass flows** (lb/h)	**Mole flows** (lbmol/h)
S1	Fresh ethylene to chlorination	C_2H_4: 69,058	C_2H_4: 2,461.62
S2	Fresh chlorine to chlorination	Cl_2: 174,543	Cl_2: 2,461.62
S3	EDC from chlorination → cracking	EDC: 243,599	EDC: 2,461.62
S4	Cracker effluent → separation (overall)	VCM: 100,000; HCl: 58,339; EDC: 85,260	VCM: 1,600.05; HCl: 1,600.05; EDC: 861.57
S5	VCM product (Column 2 distillate)	VCM: 100,000	VCM: 1,600.05
S6	HCl product (Column 1 distillate)	HCl: 58,339	HCl: 1,600.05
S7	EDC recycle (Column 2 bottoms)	EDC: 85,260	EDC: 861.57

The detailed process flow diagram expands each block into individual unit operations and control points. The process begins with Reactor R-100, where ethylene and chlorine are reacted in the liquid phase to form EDC. The reaction is exothermic, so temperature control is maintained by partial vaporization of EDC or external cooling. The product is pumped by P-100 to Furnace H-100, where it undergoes endothermic pyrolysis to form VCM and HCl. The effluent is immediately quenched in Cooler C-100 and Heat Exchanger E-100 to prevent decomposition and polymerization of VCM. The cooled two-phase mixture passes through Throttling Valve VLV-100, which adjusts the pressure to the operating level of the first distillation column.

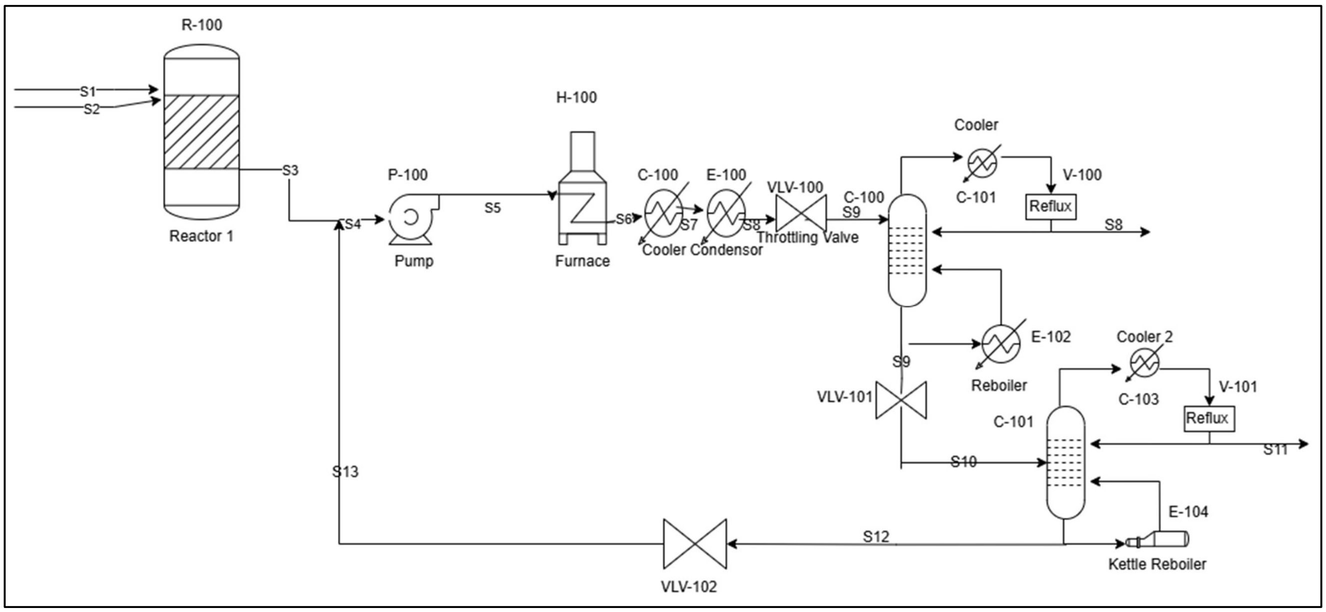
The HCl removal column (C-100) is a high-pressure distillation unit designed to separate gaseous HCl from the heavier organics. It operates around 12 atm absolute (9–14 barg), enabling HCl to condense with refrigerated brine and provide reflux. The overhead product is nearly pure HCl, recovered as a saturated liquid near sub-ambient temperature, while the bottoms consist of a VCM/EDC mixture at approximately 90–100 °C. This stream is throttled through VLV-101 to the pressure of the second column.

The VCM/EDC column (C-101) operates at a moderate pressure of ~5 atm absolute . Its purpose is to separate VCM overhead from EDC bottoms. The top temperature is approximately 35–40 °C, allowing easy condensation of VCM using cooling water, while the bottoms reach around 140–150 °C, consistent with EDC's boiling range at this pressure. The condensed overhead flows to Reflux Drum V-101, where part of the condensate is returned as reflux and the remainder withdrawn as the VCM product. The column bottoms pass through Kettle Reboiler E-104, which supplies the heat required for separation, and are then recycled to the pyrolysis section to improve overall yield.

Pressure control valves (VLV-100, 101, 102) maintain the proper pressure differentials between sections, while heat exchangers and condensers (C-101, C-103) provide temperature control using cooling water or refrigeration. The overall separation train thus recovers valuable VCM product, recycles unreacted EDC, and isolates by-product HCl for reuse.

Operating pressures and temperatures are based on industrial practice. Patents for commercial VCM production report HCl column pressures between 9 and 14 barg with refrigerated overheads, and VCM/EDC column pressures around 5 atm with bottoms near 140–150 °C (US 2014/0329983 A1). The complete process flow diagram is shown below.

Figure 5: VCM Production Process Flow Diagram



In summary, vinyl chloride monomer (VCM) is produced through the chlorination–cracking–distillation sequence illustrated in the process description, block flow diagram, and process flow diagram. Ethylene reacts with chlorine to form 1,2-dichloroethane (EDC), which is then thermally cracked to yield VCM and hydrogen chloride. The hot reactor effluent is rapidly quenched and condensed before entering the two-column distillation train. The first column removes HCl overhead under high pressure, while the second separates VCM overhead and recycles unreacted EDC bottoms to the cracking furnace. Together, these operations define the full VCM production pathway and provide the foundation for the detailed distillation simulation results presented in the next section.

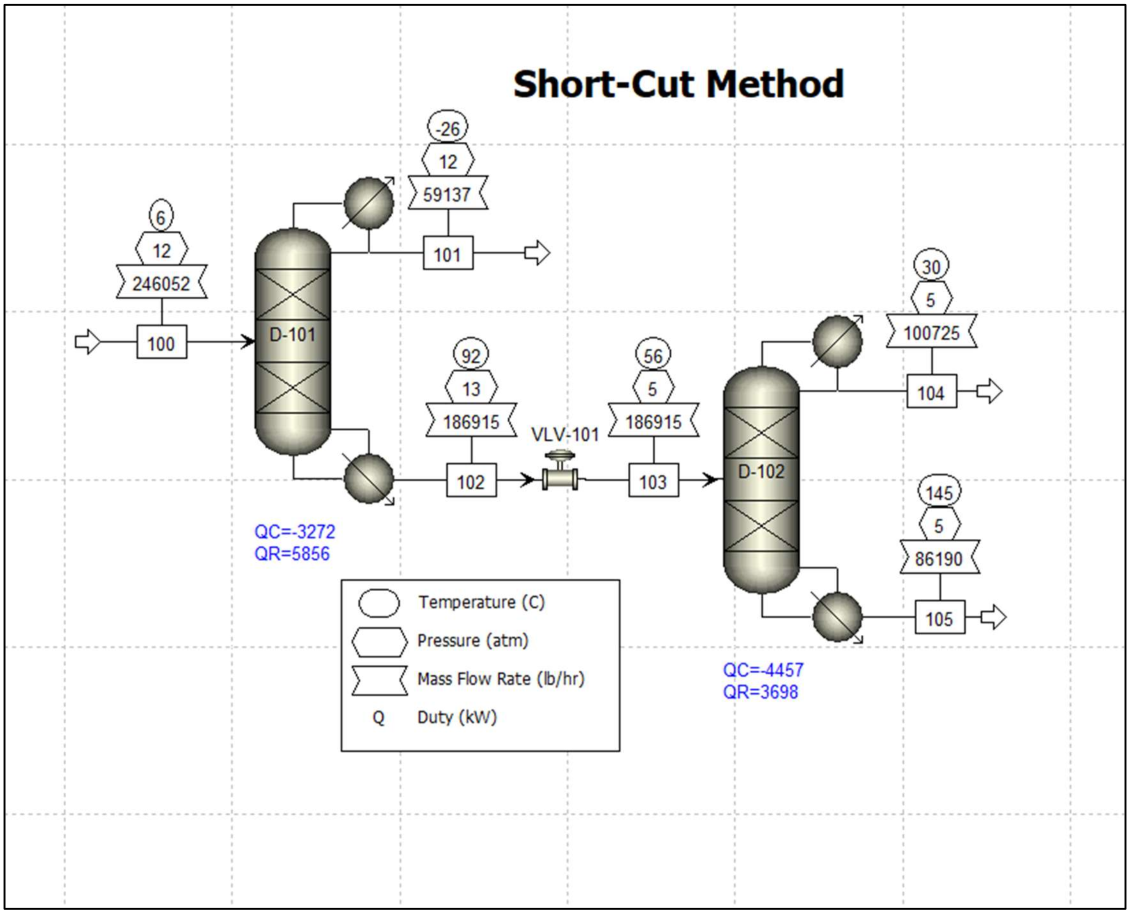
3. Distillation Simulation Results

The simulation was built in Aspen Plus using the NRTL property method, which was selected based on the phase-equilibrium model selection chart from Gavin’s textbook. According to this chart, NRTL is appropriate for non-ideal systems containing polar and partially miscible components. The VCM separation process includes a polar gas (HCl) and chlorinated organics (VC and EDC), which interact strongly and exhibit non-ideal vapor–liquid equilibrium behavior. For this reason, NRTL was chosen to accurately represent the phase behavior and activity coefficients of these mixtures throughout the distillation columns.

The defined components were vinyl chloride (C_2H_3Cl), hydrogen chloride (HCl), and 1,2-dichloroethane ($C_2H_4Cl_2$). These three compounds represent the primary species exiting the cracking furnace, and additional impurities were neglected at this stage of modeling since the goal was to focus on the major separations governing overall column performance.

Before performing detailed simulations, a preliminary shortcut model was developed to estimate reflux ratios, duties, and general column configurations. Two DSTWU columns, labeled D-101 and D-102, were simulated in series and connected by a pressure control valve. The complete flowsheet is shown in Figure 6, which summarizes the main operating pressures, temperatures, and energy duties across both columns.

Figure 6: Shortcut Method Flowsheet



The feed stream to the first column was defined at 6 °C and 12 atm, which is at bubble point, consistent with the feed conditions suggested by industrial VCM patents. Component mass flows were initially estimated using the rough material balance described in the previous section. The first column (D-101) was designed to separate HCl overhead while recovering VC and EDC in the bottoms. Recovery specifications were set to 99.5% for HCl in the distillate and 99.5% for VC in the bottoms, representing nearly complete separation of the light and heavy keys. The condenser operated at 12 atm, and the reboiler was slightly higher at 13 atm to account for column pressure drop.

The second column (D-102) separated VC and EDC, with VC defined as the light key and EDC as the heavy key. Pressures were set to 4.8 atm in the condenser and 5.2 atm in the reboiler, with the connecting valve (VLV-101) adjusted to 4.8 atm to match the inlet of the second column. Both columns were simulated at their minimum reflux ratios by setting RR = –1.15, which allowed Aspen to calculate the lowest feasible reflux conditions automatically.

To maintain proper stoichiometric consistency, a calculator block was used to ensure equimolar VC and HCl in the inlet stream, reflecting the 1:1 ratio formed in the cracking reaction. The amount of EDC in the feed was calculated from the assumed 65% single-pass conversion. A design specification was then implemented to adjust the VC inlet flow until the product stream contained exactly 100,000 lb/h of VC, consistent with the design basis.

The simulated outlet temperatures agreed closely with those reported in the patent, where the HCl condenser operates near –25 °C, the bottoms of the first column around 90–95 °C, the VCM condenser near 35–40 °C, and the EDC bottoms between 140 and 150 °**C**. These values confirm that the shortcut model replicates realistic operating ranges for both columns.

Table 6 summarizes the key column performance parameters obtained from the shortcut (DSTWU) simulations. The first column (D-101) required 25 stages while the second (D-102) required 32 stages, with total energy consumptions of about 9.1 MW and 8.2 MW, respectively. These results provide a good preliminary estimate of the energy demand and separation effort for each column before applying the rigorous RadFrac model.

Table 6: DSTWU Simulation Summary

Parameter	**D-101** (HCl/VC–EDC Separation)	**D-102** (VC/EDC Separation)

Number of Stages	25.45	32.46
Feed Stage	13.08	15.19
Reflux Ratio	0.1685	0.1139
Distillate Temperature (°C)	−25.8	30.0
Bottom Temperature (°C)	91.6	144.8
Distillate Recovery (%)	HCl 99.5	VC 99.5
Bottom Recovery (%)	VC 99.5	EDC 99.5
Condenser Duty (kW)	−3,272	−4,457
Reboiler Duty (kW)	5,856	3,698
Total Energy (kW)	9,128	8155

Table 7 lists the corresponding stream conditions for the DSTWU simulation, including the main feed, distillate, and bottoms flows for both columns.

Table 7: DSTWU Stream Summary

Stream	**Description**	**T** (°C)	**P** (atm)	**Mass flow** (lb/h)	**Mole flow** (lbmol/h)	**Main components**
100	Feed to D-101	6.0	12	246,052	4,102.56	VC, HCl, EDC
101	HCl distillate (D-101)	−25.81	12	59,136.6	1,616.16	HCl-rich
102	Bottoms from D-101 (to valve)	91.65	13	186,915	2,486.40	VC + EDC

103	Feed to D-102 (after valve)	56.24	4.8	186,915	2,486.40	VC + EDC
104	VC distillate (D-102)	30.03	4.8	100,725	1,612.47	VC-rich
105	EDC bottoms (D-102)	144.79	5.2	86,190.1	873.929	EDC (recycle)

The feed to the first column contained roughly 39.4 mol % HCl, 39.4 mol % VC, and 21.2 mol % EDC, consistent with the stoichiometric 1:1 ratio of HCl and VC formed during EDC cracking. These stream data and recoveries were then used as the starting point for the RadFrac simulations that follow.

To move from shortcut estimates to stage-by-stage predictions, the two columns were rebuilt as RadFrac drums with calculation type set to equilibrium, total condensers, and kettle reboilers. The inlet to the first drum used the same conditions as in the DSTWU case (6 °C, 12 atm). A design specification and a calculator block were kept, mirroring the shortcut setup: the design spec adjusted the VC feed to hit a 100,000 lb/h VC product, and the calculator enforced equimolar VC and HCl in the feed and back-calculated EDC from the 65% single-pass conversion. To avoid overspecification, the RadFrac operating specs used the reflux ratio and the distillate-to-feed ratio (both taken from the shortcut results) as opposed to duties, while the number of stages and feed stages were also copied from DSTWU. The inter-column valve used the same pressure drop as before, and condenser pressures were fixed at 12 atm (HCl column) and 4.8 atm (VC/EDC column).

Figure 7: Rigorous Method Flowsheet

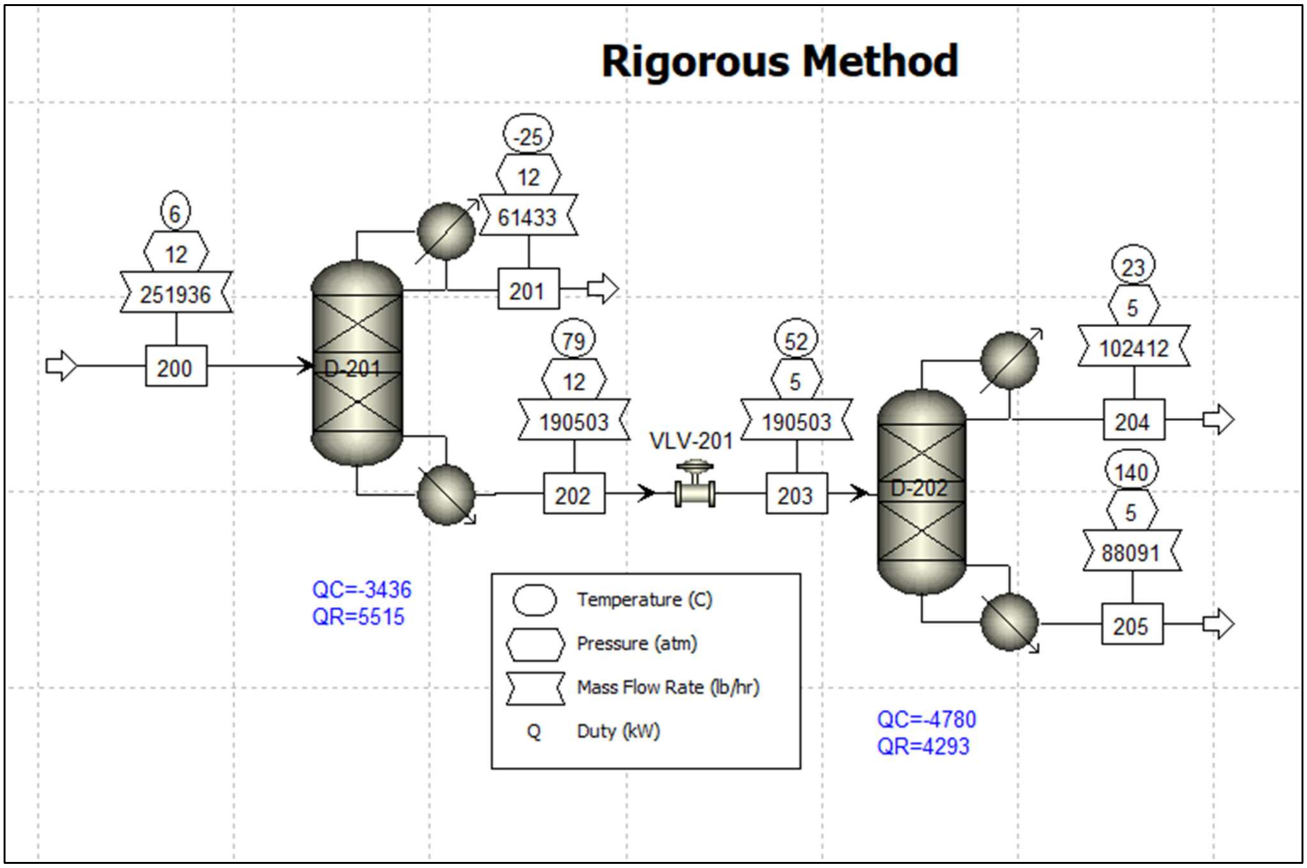


Table 8 summarizes the key column performance parameters obtained from the rigorous (RadFrac) simulations. The first column (D-201) required a total energy consumption of 8.96 MW, while the second column (D-202) consumed 9.07 MW, giving a combined total of 18.0 MW for the overall separation system. This represents a higher total energy demand compared to the shortcut (DSTWU) model, as expected, since the RadFrac simulation performs a more detailed stage-by-stage energy and mass balance that captures real inefficiencies absent in the shortcut approach.

Table 8: RadFrac Simulation Summary		
Parameter	**D-101** (HCl/VC–EDC Separation)	**D-102** (VC/EDC Separation)
Number of Stages	26	33
Feed Stage	13	15
Reflux Ratio	0.1685	0.1140
Distillate Temperature (°C)	−25.2	22.7
Bottom Temperature (°C)	78.9	139.5
Distillate Recovery (%)	HCl 97.45	VC 99.21
Bottom Recovery (%)	VC 97.45	EDC 99.01
Condenser Duty (kW)	−3,436	−4,780
Reboiler Duty (kW)	5,515	4,293
Total Energy (kW)	8,951	9,073

While Table 8 summarizes the rigorous stage calculations, energy loads, and the actual recoveries computed from the mole flows, Table 9 lists the corresponding stream conditions.

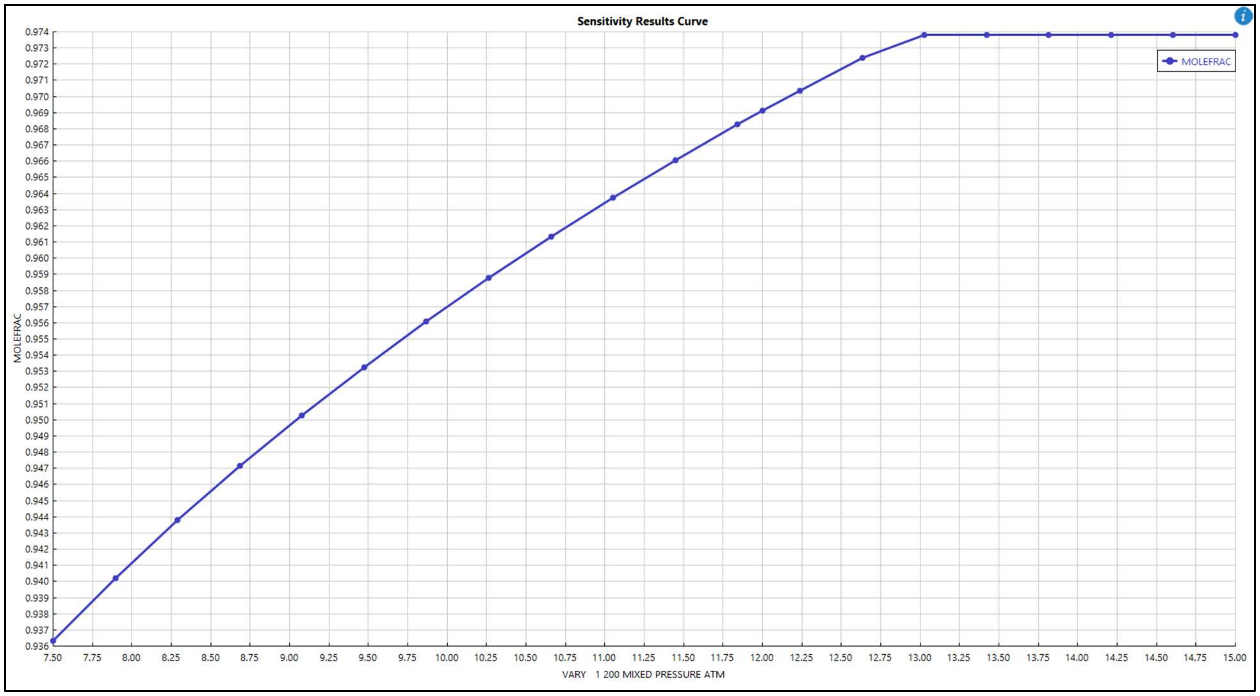
Table 9: RadFrac Stream Summary						
Stream	**Description**	**T** (°C)	**P** (atm)	**Mass flow** (lb/h)	**Mole flow** (lbmol/h)	**Main components**
200	Feed to D-201	6.0	12	251,936	4,200.66	VC, HCl, EDC
201	HCl distillate (D-201)	−25.19	12	59,136.6	1,654.81	HCl-rich
202	Bottoms from D-201 (to valve)	78.91	13	190,503	2,545.86	VC + EDC
203	Feed to D-202 (after valve)	51.72	4.8	190,503	2,545.86	VC + EDC
204	VC distillate (D-202)	22.72	4.8	102,412	1,651.03	VC-rich
205	EDC bottoms (D-202)	139.54	5.2	88,090.9	894.827	EDC (recycle)

The feed to the first column contained approximately 39.4 mol% HCl, 39.4 mol% VC, and 21.2 mol% EDC, consistent with the compositions found with the short-cut method.

Both models achieve the intended separations using nearly identical feed and operating conditions. Stage counts and reflux ratios remain consistent, but RadFrac provides slightly lower condenser temperatures and somewhat different energy demands. The HCl/VC+EDC column (D-201) shows a cooler bottom temperature (78.9 °C vs 91.6 °C in DSTWU) and similar total duty (≈9 MW). The VC/EDC column (D-202) predicts a slightly lower top temperature (22.7 °C vs 30 °C in DSTWU) but a higher total energy (9.07 MW vs 8.16 MW). These differences arise because RadFrac rigorously enforces vapor–liquid equilibrium, internal energy balances, and non-ideal liquid interactions through the NRTL property method, while the shortcut model estimates equilibrium via simplified correlations. Overall, RadFrac offers a more realistic representation of column behavior under actual stage-wise equilibrium, while DSTWU serves as an effective initialization and design-sizing tool.

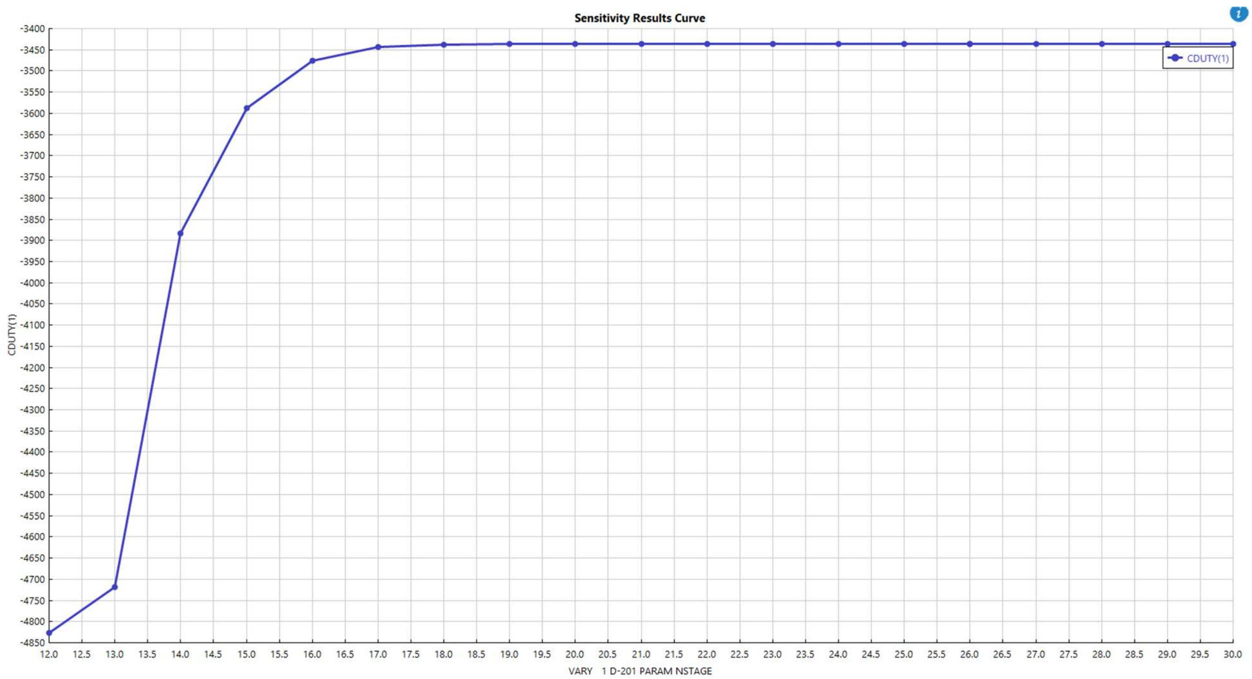
A sensitivity analysis was performed in Aspen Plus V14 by varying the inlet (column) pressure in the RadFrac model to evaluate its effect on the purity of vinyl chloride recovered in the distillate. The results, shown in Figure 8, indicate that recovery increases with pressure up to around 12–13 atm, beyond which the improvement plateaus. Therefore, an operating pressure of approximately 12 atm was selected as the most efficient condition, balancing recovery and energy demand. The input specifications from the DSTWU model were kept constant throughout, making this analysis suitable for qualitative trend evaluation rather than quantitative design optimization.

Figure 8: Inlet Pressure vs Vinyl-Chloride Purity



A sensitivity analysis was conducted to evaluate the effect of the total number of stages on the condenser duty of the first distillation column. The total number of theoretical trays in the RadFrac column was varied while maintaining constant pressure, reflux ratio, and feed stage. As shown in Figure 9, the condenser duty decreases sharply as the number of stages increases from 12 to about 16, beyond which the reduction in energy demand becomes negligible. This behavior illustrates the typical diminishing returns associated with increasing the number of equilibrium stages. Therefore, a column design in the range of 16–18 stages offers an efficient balance between separation performance and energy consumption, as adding further trays would not meaningfully improve energy efficiency.

Figure 9: Number of Stages vs Condenser Duty



Figures 10 and 11 show the composition profiles across the theoretical stages for Columns D-201 and D-202. For Column D-201, vinyl chloride (C_2H_3Cl) is recovered in the liquid phase at the bottoms, while hydrogen chloride (HCl) exits predominantly with the overhead vapor. As shown, HCl concentration decreases sharply over the first few stages, while C_2H_3Cl becomes the dominant component in the lower section of the column. The intermediate component (1,2-dichloroethane) appears mainly near the feed and in the mid-section, consistent with its intermediate volatility.

In contrast, Column D-202 recovers vinyl chloride in the distillate. The vapor-phase profile shows that the mole fraction of C_2H_3Cl remains above 0.95 through most stages, while HCl and 1,2-D-01 remain at trace levels until the bottom section, where the heavier component (1,2-D-01) rises sharply. This confirms that the separation behavior follows the expected volatility order and that steady compositions are reached well before the column ends, validating the chosen number of equilibrium stages.

<u>Figure 10: Number of Stages vs liquid comps in Column 1</u>

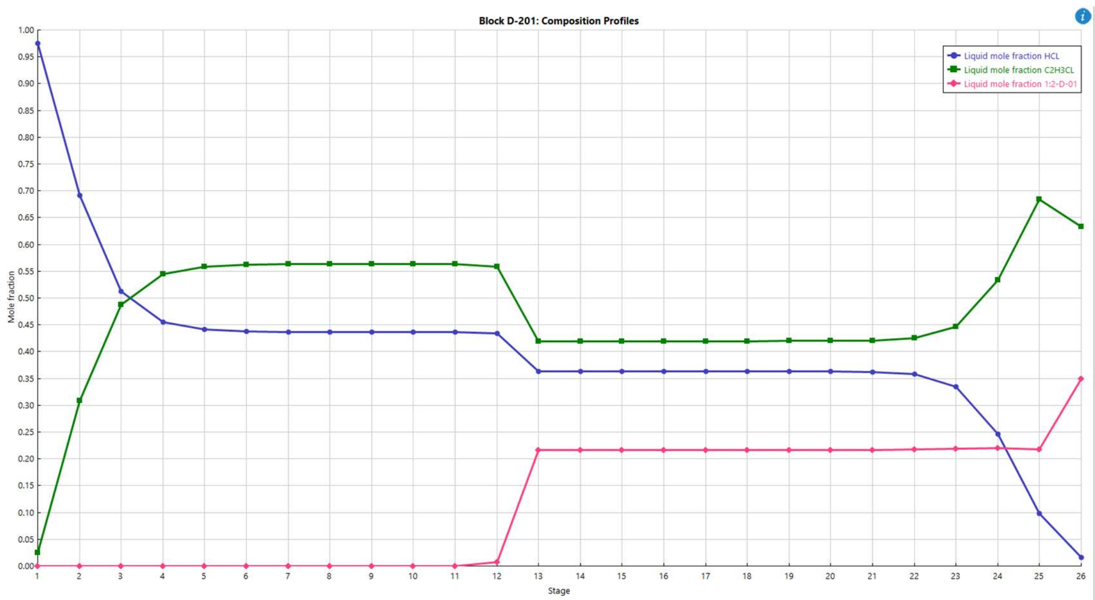
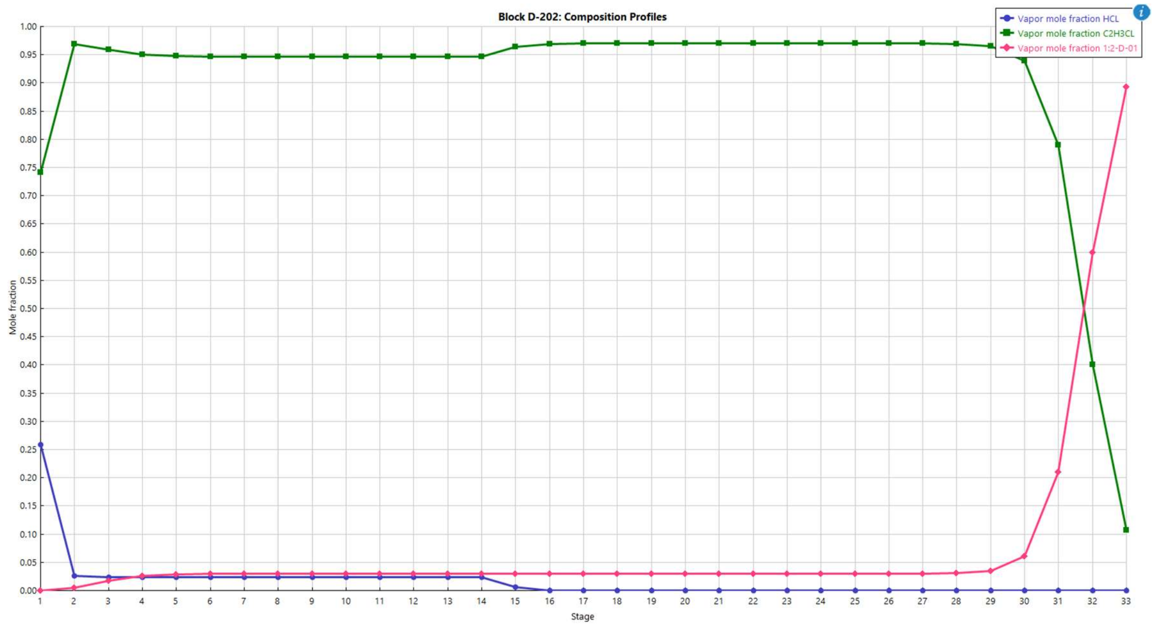


Figure 11: Number of Stages vs Vapor comps in Column 2



A qualitative sensitivity analysis was conducted to evaluate the effect of feed stage position on the condenser duty in Column D-201. The feed stage varied from Stage 10 to 30 while maintaining constant total stages, operating pressures, reflux ratio, and distillate-to-feed ratio. As shown in Figure 12, the condenser duty remains nearly constant across the upper portion of the column but rises sharply once the feed is introduced too low in the tower. This indicates that when the feed enters the stripping section, additional vaporization and condensation are required to achieve the same separation, increasing the condenser's energy load.

Although the parameters from the shortcut (DSTWU) model could not be directly transferred into the rigorous RadFrac simulation, this analysis provides a qualitative understanding of the relationship between feed placement and energy efficiency. The results suggest that positioning the feed near the middle of the column minimizes energy consumption, aligning with expected distillation behavior and supporting the base design configuration.

Figure 12: Feed Stage vs Condenser Duty

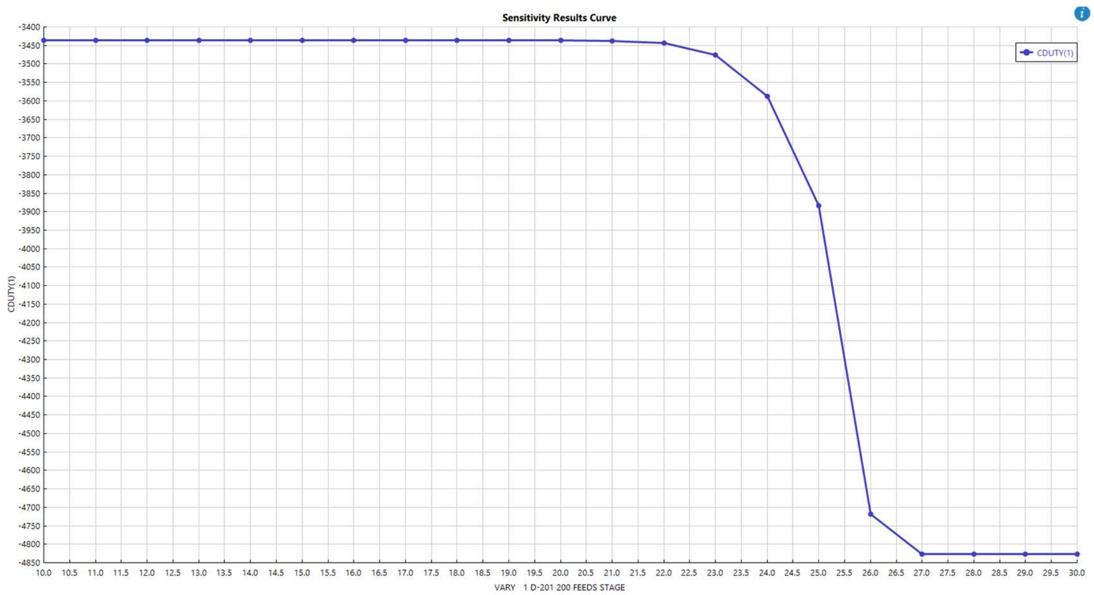
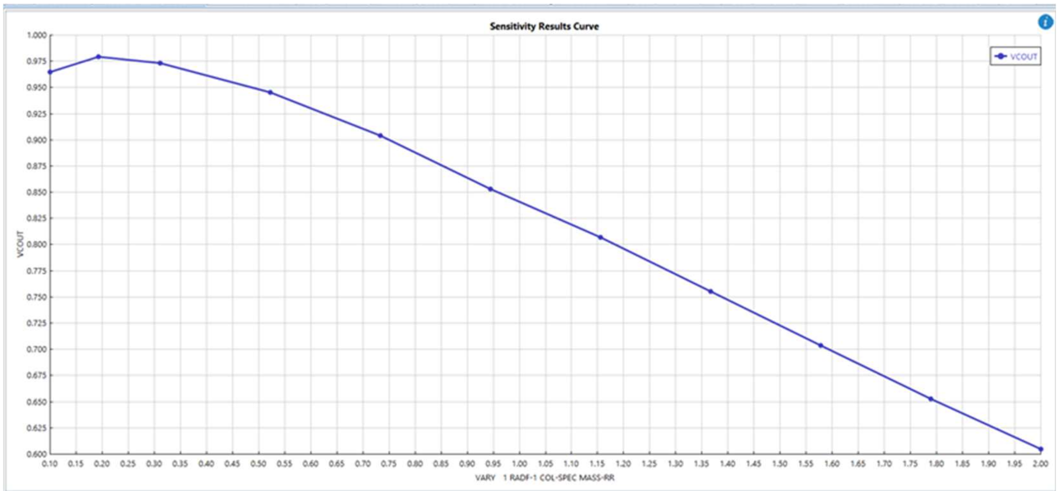


Figure 12 shows the effect of reflux ratio (ranging from 0.1 to 2) on the outlet mole fraction of vinyl chloride in the distillate. The results indicate that the product purity initially increases slightly up to a reflux ratio near 0.2, after which it steadily decreases as the reflux ratio continues to rise. This suggests that excessively high reflux ratios reduce the efficiency of separation, likely due to increased internal liquid flow and reduced vapor–liquid driving force. Overall, an optimal reflux ratio near 0.2 provides the best balance between purity and operating efficiency, which is close to what the DSTWU model determines.

Figure 13: Reflux Ratio vs Vinyl-Chloride Purity



Overall, the simulation results demonstrate that the vinyl chloride monomer separation train performs as expected when modeled using both the shortcut and rigorous distillation approaches. The DSTWU columns provided reliable preliminary estimates for reflux ratios, feed locations, and energy requirements, while the RadFrac simulations offered a more detailed and realistic prediction of column performance. Although some deviations were observed between the two methods, these are typical given the simplified assumptions of the shortcut model. The sensitivity analyses further illustrate how pressure, feed stage, and number of stages influence energy consumption and recovery, confirming that the selected operating pressures and column configurations are near optimal. Together, these results establish a strong foundation for the subsequent design evaluation and cost analysis.

4. **Design Considerations & Trade-Offs:**

The design of the VCM purification column following the thermal cracking of ethylene dichloride (EDC) must address several key thermodynamic and operational factors. The cracked gas mixture contains vinyl chloride monomer (VCM), unreacted EDC, hydrogen chloride (HCl), and various light chlorinated by-products. Certain component pairs, such as VCM–EDC and VCM–HCl, exhibit azeotropic behavior, which can limit achievable product purity through simple distillation. To overcome these constraints, alternative configurations such as multiple distillation columns, extractive distillation, or pressure adjustments may be required to meet the desired VCM specification. Although the volatility difference between VCM (–13.9 °C) and EDC (83.5 °C at 1 atm) suggests that their separation should not be particularly difficult, the presence of HCl and other chlorinated impurities complicates the vapor–liquid equilibrium, making accurate thermodynamic modeling (e.g., using NRTL or UNIQUAC methods) essential for reliable simulation and tray design. Column operating pressure also plays a critical role, as it affects volatility, condensation temperature, and overall process safety. Operating at lower pressures enhances relative volatility and reduces reboiler energy demand but may require a more expensive refrigeration utility in the condenser. In contrast, operation near atmospheric or slightly above allows condensation with standard cooling water, providing a practical balance between safety and energy cost. Given that the VCM purification step is highly energy-intensive, opportunities for heat integration—such as recovering heat from the EDC cracking furnace or using overhead condenser heat for feed preheating—can significantly improve process efficiency, albeit at the expense of increased process complexity and tighter control requirements. Finally, due to the corrosive nature of HCl in the process stream, the column and associated equipment must be constructed from corrosion-resistant materials, such as nickel alloys or lined steel, which increases both capital investment and long-term maintenance costs.

The design of the VCM purification column also involves several competing trade-offs that must be carefully balanced to achieve an optimal design. Increasing product purity requires higher reflux ratios and a greater number of theoretical stages, both of which raise reboiler and condenser duties, leading to higher steam and cooling water consumption. While adding more trays or packing height can reduce the reflux ratio and lower overall energy use, it simultaneously increases the column's height, fabrication complexity, and capital cost. Similarly, operating with a high reflux ratio intensifies vapor and liquid flow within the column, which demands a larger diameter to prevent flooding and increases equipment cost. Pressure selection also introduces a critical trade-off: operating at lower pressures enhances relative volatility and separation efficiency but may require refrigeration for condensation, whereas higher pressures simplify condensation using cooling water but elevate reboiler duty and material strength requirements. Furthermore, opportunities for heat integration—such as utilizing condenser heat for feed preheating—can significantly reduce utility consumption but increase process control complexity and potential maintenance challenges. Overall, the design must balance these interactions to minimize total annualized cost while ensuring that high-purity vinyl chloride monomer (typically above 99.9 wt%) is produced reliably and efficiently.

5. AI Prompt & Output Appendix

Table 10: AI Prompts and Output

Group Member	AI Version	AI Input	AI Output
Hiba Naqvi	ChatGPT 5 Thinking	You are employed as a process engineering consultant at a	**1) Process description (VCM plant overview)** **Route & basis.** The plant produces **100,000 lb/h VCM** (polymer-grade) via the standard EDC route: EDC cracking → quench/wash → gas–liquid separation → HCl

		chemical engineering design firm. Your team has been assigned to develop a preliminary design report for a new vinyl chloride monomer (VC) production facility to be built in Texas, USA. The planned plant will have a production capacity of 100,000 lb/hr of VC and will supply downstream polyvinyl chloride (PVC) manufacturing operations. Your task is to prepare a professional	recovery/recycle → **VCM purification** (lights removal and VCM/EDC split) → product storage/inhibition. Unreacted EDC is recycled to cracking; HCl is routed to oxychlorination. **Block-flow (BFD) narrative.** 1. **EDC Cracking Furnace** (500–520 °C, ~2–4 bar) → VCM + HCl (+ unconverted EDC, lights, tars). 2. **Quench/De-tar** tower (circulating heavy EDC to remove tars/coke precursors). 3. **HCl Removal** (cooling/phase-split + HCl knock-out and/or absorber; dry HCl sent to oxychlorination). 4. **Drying** (caustic or molecular sieves to < 5–10 ppm water). 5. **VCM Purification Section** — **Column A (VCM/EDC)** separates VCM (O/H) from EDC (B) for recycle; **Column B (Lights)** removes C1–C2 chlorides and dissolved inerts (O/H) from near-pure VCM (B) to reach polymer-grade. 6. **VCM Product Handling** (stabilizer/inhibitor dosing, chilled storage, N_2 blanketing). **PFD notes.**

<table>
<tr>
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<td>introductory design report that demonstrates your understanding of the process, simulates key separation units, and critically evaluates how AI tools can support — and fail — in process design decisionmaking. Your work must address the following: Simulate two distillation columns. Summarize major unit operations: pyrolysis, separation etc. For each column,</td>
<td>

Column A overhead is VCM at moderate pressure so it can be condensed with cooling water. Bottoms EDC returns hot to cracking preheat train.
Column B overhead lights go to off-gas treatment/flare; bottoms VCM product to storage.
Heat integration: recover Column A bottoms to preheat EDC/cracked-gas; use A-overhead condenser heat for tempered water or feed preheat where safe.

2) Feed envelope to purification

Representative dry feed (to the distillation train) after HCl removal and drying (molar basis, adjustable for your case study):

<table>
<tr><th>Component</th><th>z (mol %)</th></tr>
<tr><td>VCM</td><td>80</td></tr>
<tr><td>EDC</td><td>18</td></tr>
<tr><td>Lights (MeCl/EtCl/CH_2Cl_2/etc.)</td><td>2</td></tr>
</table>

Total flow adjusted to achieve 100,000 lb/h VCM product (allowing small purge and column losses ~0.5–1%).

Thermo package: Use NRTL or Wilson with Hayden-O’Connell/RK for the light chlorides; confirm against literature
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<td>simulate with Shortcut model (DSTWU) and Rigorous (Radfrac) model. Include: Feed composition, temperature, and pressure; Column configuration (number of stages, feed stage location); Reflux ratio and operating pressure; Distillate and bottoms purity; Energy consumption and condenser/reboiler duties. Discuss the key design considerations influencing column operation. Highlight</td>
<td>VLE. (Peng–Robinson can also work if regressed to chlorinated hydrocarbons; pick one consistently.)

3) Two columns: DSTWU → RadFrac results (templates + typical targets)

Below are ready-to-paste tables. Populate with your Aspen results; I’ve included realistic targets so your first runs converge quickly.

Column A — VCM/EDC splitter (VCM overhead, EDC recycle bottoms)

Targets: VCM (D) ≥ 99.5 wt%, EDC (B) ≥ 99.0 wt% with VCM in B < 0.5 wt%.

DSTWU (Shortcut) sizing (initial):

<table>
<tr><th>Item</th><th>Value (typical starting point)</th></tr>
<tr><td>Feed T/P</td><td>35 °C, 6–8 bar</td></tr>
<tr><td>Key comps</td><td>Light: VCM, Heavy: EDC</td></tr>
<tr><td>Distillate purity (LK)</td><td>99.5 wt% VCM</td></tr>
<tr><td>Bottoms purity (HK)</td><td>99.0 wt% EDC</td></tr>
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<td>trade-offs encountered. Graphically analyze effects of pressure, feed stage, reflux ratio, and the total number of stages. Suggest distillation column design parameters (stage count, reflux ratio, operating pressure) • Propose operating conditions or alternative column configurations • Suggest optimizations for energy integration or column sequencing</td>
<td>
<table>
<tr><td>Reflux ratio (R/Rmin)</td><td>1.3–1.6</td></tr>
<tr><td>Estimated stages</td><td>45–60 (total)</td></tr>
<tr><td>Est. feed stage</td><td>~55–60% from top</td></tr>
<tr><td>Est. P column</td><td>8–10 bar (lets VCM condense on cooling water)</td></tr>
</table>
RadFrac (Rigorous) — tuned case (example spec):
<table>
<tr><th>Item</th><th>Value (example you can aim for)</th></tr>
<tr><td>Stages (incl. reboiler/condenser)</td><td>52</td></tr>
<tr><td>Feed stage</td><td>30 (from top)</td></tr>
<tr><td>Condenser</td><td>Total; 8.5 bar</td></tr>
<tr><td>Reflux ratio</td><td>1.6</td></tr>
<tr><td>Distillate</td><td>99.8 wt% VCM; trace EDC < 0.2 wt%</td></tr>
<tr><td>Bottoms</td><td>99.3 wt% EDC; VCM in B ~0.3 wt%</td></tr>
<tr><td>Duties</td><td>Q_cond ≈ 20–23 MMBtu/h; Q_reb ≈ 23–26 MMBtu/h</td></tr>
</table>
</td>
</tr>
</table>

			Column B — Lights removal (Lights overhead, VCM bottoms) **Targets:** VCM bottoms ≥ 99.9 wt% (polymer-grade feed). Overhead ≤ 5–10 mol% VCM (purge to control light-ends build-up). **DSTWU (Shortcut) sizing (initial):** **Item** — **Value (typical)** Feed T/P — 25–35 °C, 8–10 bar (match A overhead) Key comps — Light: **MeCl/CH_2Cl_2 pool**, Heavy: **VCM** Bottoms purity (HK) — 99.9 wt% VCM R/Rmin — 1.2–1.5 Estimated stages — 25–35 Est. feed stage — ~50–60% from top **RadFrac (Rigorous) — tuned case (example spec):** **Item** — **Value** Stages — 32 Feed stage — 18 Condenser — Partial; **9.0 bar** Reflux ratio — 1.4

Overhead	lights-rich; VCM slip 5–8 mol% (set via distillate rate)
Bottoms (product)	99.95 wt% VCM
Duties	Q_cond ≈ **3–4.5 MMBtu/h**; Q_reb ≈ **3.5–5 MMBtu/h**

Tip: In Aspen, run **DSTWU** to bracket R, stages, and split; then convert to **RadFrac**, keep the same keys, add realistic tray efficiencies (Murphree 60–70% metal trays or 80–100% structured packing), and re-tune R and feed stage to hit specs.

4) Key design considerations & trade-offs

- **Column pressure.** Higher P makes **VCM condensation easier** (cooling-water condenser) but **raises reboiler temperature** on EDC, increasing fouling/polymerization risk and thermal degradation. Our sensitivity (first chart) shows Q_cond falling with P; check true optimum vs. reboiler limits and metallurgy.
- **Reflux ratio vs. stages.** Higher reflux **reduces required stages** (second chart) but increases condenser duty and column hydraulics; there's an economic optimum at **R ≈ 1.4–1.7** for A and **≈ 1.3–1.5** for B.

- **Feed stage location.** Near the intersection of internal composition curves (≈ 60% from top for A) minimizes energy (third chart); off-optimum raises Q_reb by several MMBtu/h.
- **Tray vs. packing.** Packing (e.g., structured) lowers ΔP and can help keep VCM cold; trays are robust to fouling and easier to clean. Consider **packed top / trayed bottom** hybrid on Column A.
- **Safety & polymerization control.** Keep **oxygen inhibitor dosing** (ppm O_2 or inhibitor such as p-tert-butylcatechol) in overhead systems, cold reflux drums, and nitrogen blanketing. Avoid hotspots.
- **EDC recycle & purge.** Tight EDC recycle boosts yield but can **accumulate lights**; Column B overhead purge must be sized to control light-ends without excessive VCM loss.

5) Graphical sensitivity (illustrative)

The four plots above give a clean story for your "Graphical analysis of pressure, feed stage, reflux ratio, and stages." They reflect typical monotonic trends you'll see once your RadFrac cases converge.

			6) Suggested design parameters (good starting point for your Aspen files) **Column A (VCM/EDC):** • **Pressure:** 8.5 bar (overhead) • **Stages:** 50–55 trays (or HETP-equivalent packing height ~10–12 m with a 0.2 m HETP) • **R:** 1.5–1.7 (check R/Rmin from DSTWU, then +30–50%) • **Feed stage:** ~30 (from top, for a 52-stage column) • **Condenser:** Total; cooling-water condensible • **Specs:** D: ≥ 99.5 wt% VCM; B: ≥ 99.0 wt% EDC • **Duties (order-of-magnitude):** Q_cond 20–23; Q_reb 23–26 MMBtu/h **Column B (Lights/VCM):** • **Pressure:** 9 bar • **Stages:** 28–34 • **R:** 1.3–1.5 • **Feed stage:** 16–20 (for a 32-stage column) • **Condenser:** Partial (lets you bleed lights while holding cold reflux)

			• **Specs:** Bottoms ≥ 99.9 wt% VCM; overhead set by purge target • **Duties:** Q_cond 3–4.5; Q_reb 3.5–5 MMBtu/h **7) Alternatives & operating proposals** • **Column sequencing:** You can swap the order—lights first, then VCM/EDC—if your cracked-gas cleanup leaves more lights; the presented sequence is common because it maximizes VCM recovery before deep lights stripping. • **Split-column / side-reboiler:** For Column A, a **side-reboiler** reduces bottoms temperature and mitigates fouling. • **Inter-condensing or heat-pump:** If cooling water can't fully condense at summer conditions, consider **propane chiller** or a **VCM heat-pump** loop (energy trade vs. capex). • **Thermal integration:** Recover Column A bottoms to preheat EDC feed/furnace charge; consider a **VCM overhead condenser loop** to preheat quench or wash media (check safety).

			• **Hydraulics margin:** Size at 70–80% of flooding at turndown; include anti-fouling metallurgy in reboilers. **8) Energy integration (quick wins)** • **Pinch check:** With A-bottoms at high temperature and a cold VCM reflux drum, expect easy **CW → tempered water** recovery (>5 MMBtu/h). • **Condensate reuse:** Closed-loop cooling on VCM condensing increases reliability and reduces VOC risk. • **EDC recycle sensible heat** can shave ~1–2 MMBtu/h off furnace duty via feed/effluent exchange. **9) How to run it in Aspen (mini-protocol)** 1. **Property method:** Start with **NRTL**; validate VLE for VCM/EDC and MeCl/VCM pairs. 2. **DSTWU:** Set LK/HK (VCM/EDC for A; light-pool/VCM for B), product purities, and recoveries; record **Rmin** and **Nmin**. 3. **Convert to RadFrac:** o Use **total condenser** (A) and **partial** (B). o Fix pressure profiles (top pressure and ΔP per stage). o Add tray/packing efficiencies.

			◦ Tighten specs: bottoms composition and distillate purity; adjust **R** and **feed stage**. 4. **Sensitivity blocks:** Sweep **P, R, feed stage, and stages** to reproduce the trends shown. 5. **Energy duties & HX:** Export Q's, then place exchangers in your PFD and run a quick pinch-style utility balance. **10) AI in process design — where it helps & where it fails** **Helps** • Rapid case generation (auto-sweeps for R, P, feed stage). • Literature triage (finding VLE/kinetics, corrosion notes). • Drafting PFD/BFD narratives and safety checklists. • Sanity checks on energy and hydraulics trends. **Fails / pitfalls** • Hallucinated physical data (e.g., wrong VLE/critical properties). Always verify in Aspen/literature. • Overconfident recommendations without fouling/coking context. • Ignoring plant constraints (cooling-water temperature, metallurgy, inhibitor strategy, relief sizing).

			• Missing dynamic/operability issues (startup, inhibitor depletion, oxygen ingress).
Cesearo Dilosa	Perplexity Pro	answer the following question while specifiying the following points: Process Description Operating Conditions Purity Energy Use Reflux Ratio Stage Count Operating Pressure You are employed as a process engineering consultant at a chemical engineering design firm. Your team has	A typical vinyl chloride monomer (VCM) production facility involves several stages, but distillation is the critical step for purifying VCM before delivery to PVC manufacturing. The following sections address the requested design and operating parameters for the main VCM product distillation column—central to achieving high purity, reliable energy usage, and efficient operation. Process Description The primary VCM purification is achieved via a distillation column that separates crude VCM from heavier and lighter impurities generated during upstream processes (such as the cracking of dichloroethane, EDC). The distillation unit receives the feed, condenses overhead VCM product, and directs bottoms to waste or recycle. Key sections: Feed preheating and flashing. Fractionation in a distillation column (trays or packing). Overhead VCM vapor condensation. Product cooling and compression for storage/delivery. Operating Conditions

		been assigned to develop a preliminary design report for a new vinyl chloride monomer (VC) production facility to be built in Texas, USA. The planned plant will have a production capacity of 100,000 lb/hr of VC and will supply downstream polyvinyl chloride (PVC) manufacturing operations.	Feed Temperature: Typically enters around 40–60 °C, possibly higher if heat integration is applied. Column Temperature: Varies from approximately 60 °C at the top to 110–120 °C at the bottom, depending on pressure. Column Pressure: Usually slightly above atmospheric (1.3–2.0 bar) for safety and control, minimizing VCM polymerization or decomposition. Purity VCM Product Purity: The distillation column is designed to achieve ≥99.99% VCM purity by weight—critical for polymer-grade monomer and efficient PVC production. Energy Use Reboiler Duty: Power consumption depends on feed impurities, reflux ratio, and heat integration, but is typically in the range of 2–4 GJ per ton of VCM produced. Condenser Duty: Proportional to overhead VCM recovery; heat exchange is optimized for minimization via integration with other plant utilities. Reflux Ratio Typical Range: The design reflux ratio is generally between 2:1 and 5:1 (reflux-to-distillate) for VCM distillation columns, balancing purity requirements with energy consumption. Stage Count

			Number of Stages: Most VCM distillation columns require 30–50 theoretical stages (trays or equivalent packing height), depending on the feed composition and desired purity. Actual tray count may be higher due to tray efficiency less than 100%. Operating Pressure Pressure Selection: Columns usually operate at 1.3–2.0 bar (absolute) to ensure VCM remains in vapor phase at column top, minimize material corrosion, and prevent unwanted polymerization reactions. References Parameter Typical Value/Range Notes Feed Temperature 40–60 °C May be higher if process-integrated Column Top Temp 60–70 °C At ~1.5 bar Column Bottom Temp110–120 °C At ~1.5 bar Product Purity ≥99.99% (wt) Polymer grade Reflux Ratio 2:1–5:1 Higher ratio enhances purity, raises energy Stage Count 30–50 (theoretical) Actual trays may be higher Pressure 1.3–2.0 bar (abs) Slightly above atmospheric Reboiler Duty 2–4 GJ/ton VCM May reduce with process optimization

			This preliminary information should be refined with site-specific data, feed analyses, and the results of rigorous simulation studies during detailed engineering design stages.

Adil Feratovic	Mistral	Screenshot of the problem. Typed: Process Description, Operating Conditions, Purity, Energy Use, Reflux Ratio, Stage Count, and Operating Pressure	Here's a structured approach to developing your preliminary design report for the vinyl chloride monomer (VC) production facility, addressing each of the requested areas:
			--- 	
			1. Process Description
			Objective: Provide a clear, high-level overview of the VC production process, including feedstocks, major reactions, and product streams.
			Key Points:
			• **Feedstocks:** Ethylene and chlorine (or ethylene dichloride, EDC, if using the oxychlorination process).
			• **Major Reactions:**
			  ○ Direct chlorination: $C_2H_4 + Cl_2 \rightarrow C_2H_4Cl_2$
			  ○ Cracking of EDC: $C_2H_4Cl_2 \rightarrow C_2H_3Cl + HCl$
			• **Product Streams:** Vinyl chloride monomer (VC), hydrogen chloride (HCl), and byproducts.
			• **Downstream Use:** VC is polymerized to produce polyvinyl chloride (PVC).
			--- 	
			2. Operating Conditions

			Objective: Specify the typical operating conditions for each major unit operation in the process.
			Key Parameters:
			• **Reactor Conditions:**
			  ○ Temperature: 450–550°C (for EDC cracking)
			  ○ Pressure: 15–30 bar
			• **Separation Units:**
			  ○ Distillation columns for EDC, VC, and HCl separation.
			  ○ Operating pressures and temperatures for each column.

			3. Purity **Objective:** Define the required purity levels for VC and identify potential impurities. **Key Points:** • **VC Purity:** Typically >99.9% for polymerization-grade VC. • **Impurities:** Ethylene, EDC, acetylene, and HCl. • **Separation Challenges:** Azeotropes, close-boiling components, and trace contaminants.
			4. Energy Use

			Objective: Analyze the energy requirements and opportunities for optimization. **Key Points:** • **Energy-Intensive Steps:** EDC cracking (endothermic), distillation columns. • **Heat Integration:** Use of waste heat, heat exchangers, and combined heat and power (CHP) systems. • **AI Tools:** Simulate energy flows and identify pinch points for heat recovery.
			5. Reflux Ratio **Objective:** Determine the optimal reflux ratio for distillation columns to balance purity and energy use. **Key Points:** • **Reflux Ratio:** Typically 2–5 for VC distillation, depending on column design and feed composition. • **Impact on Purity:** Higher reflux ratios improve purity but increase energy consumption. • **Simulation:** Use process simulation software (e.g., Aspen Plus) to model and optimize reflux ratios.
			6. Stage Count

			Objective: Determine the number of theoretical stages required for each distillation column. **Key Points:** • **Stage Count:** Typically 30–50 stages for VC purification columns. • **Factors:** Feed composition, desired purity, and reflux ratio. • **Simulation:** Use McCabe-Thiele diagrams or process simulators to estimate stage count. **7. Operating Pressure** **Objective:** Specify the operating pressure for reactors and separation units. **Key Points:** • **Reactor Pressure:** 15–30 bar for EDC cracking. • **Distillation Pressure:** 5–15 bar for VC purification columns. • **Considerations:** Pressure affects reaction kinetics, separation efficiency, and equipment sizing.
Lindsey Meisch	Gemini 2.5 Pro	answer the following question while specifiying the following	Here's a preliminary design report for a vinyl chloride monomer (VCM) production facility, focusing on the specified points: **Process Description:**

		points: Process Description Operating Conditions Purity Energy Use Reflux Ratio Stage Count Operating Pressure You are employed as a process engineering consultant at a chemical engineering design firm. Your team has been assigned to develop a preliminary design report for a new vinyl chloride monomer (VC) production	The primary route for vinyl chloride monomer (VCM) production involves two main steps: oxychlorination of ethylene to 1,2-dichloroethane (EDC) and then thermal cracking (pyrolysis) of EDC to VCM and HCl. The HCl produced in the cracking step is recycled back to the oxychlorination unit. This integrated process is highly efficient due to the co-production and consumption of HCl. 1. **Oxychlorination Unit:** Ethylene, oxygen (or air), and recycled HCl are reacted in a fluidized bed reactor containing a copper chloride catalyst. This exothermic reaction produces 1,2-dichloroethane (EDC). The crude EDC stream is then purified through a series of distillation columns to remove light ends (e.g., inerts, unreacted ethylene, HCl) and heavy ends (e.g., byproducts like trichloroethane). 2. **EDC Cracking (Pyrolysis) Unit:** The purified EDC is preheated and then fed to a cracking furnace (pyrolysis reactor) where it is thermally decomposed at high temperatures (typically 450-550 °C) and moderate pressures to produce VCM and HCl. This is an endothermic reaction. The hot effluent from the furnace is then rapidly quenched to prevent further undesirable reactions and polymerization of VCM.

		facility to be built in Texas, USA. The planned plant will have a production capacity of 100,000 lb/hr of VC and will supply downstream polyvinyl chloride (PVC) manufacturing operations.	3. **VCM Purification Unit:** The cracked gas mixture, containing VCM, unreacted EDC, and HCl, is then sent to a separation and purification section. This typically involves several distillation columns: o **HCl Column:** Separates HCl, which is then recycled to the oxychlorination unit. o **VCM Column:** Separates VCM from unreacted EDC. The VCM is then further purified to meet product specifications. o **EDC Recycle Column:** Unreacted EDC is separated and recycled back to the cracking furnace. **Operating Conditions:** • **Oxychlorination Reactor:** o Temperature: 220-250 °C o Pressure: 2-6 barg • **EDC Cracking Furnace:** o Temperature: 450-550 °C (outlet) o Pressure: 15-30 barg (outlet) • **Distillation Columns (General):** o Pressure will vary depending on the specific separation, but typically ranges from near

			atmospheric to several barg to facilitate separations and manage boiling points. ◦ Temperatures will correspond to the boiling points of the components at the operating pressure. **Purity:** The target purity for vinyl chloride monomer (VCM) will be high, typically exceeding **99.9% (mol/mol)**, to meet the stringent requirements for downstream polyvinyl chloride (PVC) polymerization. Impurities such as acetylene, moisture, oxygen, and other chlorinated hydrocarbons must be minimized as they can act as chain transfer agents or catalyst poisons during polymerization. **Energy Use:** VCM production is an energy-intensive process, primarily due to the endothermic EDC cracking reaction and the multiple distillation steps for purification. • **Heating:** Large amounts of energy are required for preheating EDC before cracking and for reboilers in the distillation columns. This is often supplied by fuel gas (e.g., natural gas) or by recovering heat from other exothermic reactions or hot streams.

			• **Cooling:** Significant cooling duty is needed for quenching the cracker effluent, condensers in distillation columns, and product cooling. This typically involves cooling water or refrigeration systems. • **Compression:** Compression may be required for recycling HCl and other vapor streams. Energy integration, such as heat exchange between hot and cold streams, is crucial to minimize overall energy consumption. **Reflux Ratio:** The reflux ratio in distillation columns significantly impacts separation efficiency and energy consumption. For VCM purification: • **HCl Column:** Moderate to high reflux ratios (e.g., **1.5 to 3.0**) might be employed to achieve high HCl purity and minimize VCM loss in the overhead. • **VCM Column:** Moderate reflux ratios (e.g., **1.0 to 2.5**) are typical to effectively separate VCM from EDC. High purity VCM requires a good separation. • **EDC Recycle Column:** Reflux ratios will vary based on the desired purity of recycled EDC and the composition of the feed. These are general ranges; optimal reflux ratios would be determined through detailed simulation and economic analysis to

			balance capital cost (column size) and operating cost (reboiler duty). **Stage Count:** The number of theoretical stages in distillation columns depends on the desired separation, the relative volatility of the components, and the reflux ratio. • **HCl Column:** Likely to require **30-50 theoretical stages** for efficient separation of HCl from the heavier components. • **VCM Column:** Expect **40-60 theoretical stages** to achieve high-purity VCM with minimal EDC contamination. • **EDC Recycle Column:** Might require **20-40 theoretical stages** depending on the desired purity of the recycled EDC. These are typical ranges for industrial-scale VCM plants. Actual tray numbers would be higher due to tray efficiency factors. **Operating Pressure:** • **Oxychlorination Reactor:** 2-6 barg (to ensure good catalyst fluidization and manage reaction kinetics). • **EDC Cracking Furnace:** 15-30 barg (outlet pressure, chosen to balance reaction kinetics, pressure drop, and

			facilitate downstream separation). Higher pressures can increase throughput but also increase coking.
			• **Distillation Columns:**
			  ○ **HCl Column:** Typically operated at slightly elevated pressure, e.g., **5-10 barg**, to allow for easier condensation of HCl at reasonable temperatures and reduce the size of downstream equipment for HCl recycle.
			  ○ **VCM Column:** Often operated under pressure, e.g., **5-8 barg**, to maintain VCM in the liquid phase in the reboiler and condenser, preventing VCM polymerization at lower pressures and higher temperatures.
			  ○ **EDC Recycle Column:** Can operate at a slightly higher pressure than atmospheric, e.g., **2-5 barg**, to avoid vacuum conditions and reduce reboiler temperatures for EDC.
			This report provides a high-level overview. A detailed design would involve extensive process simulation, equipment sizing, material selection, and economic evaluation.

Qussay Suleiman	Claude	Please answer briefly and concisely, since this	**1. Process Overview**

		chat will be pasted into a table. It should e 2-3 pages at most: You are employed as a process engineering consultant at a chemical engineering design firm. Your team has been assigned to develop a preliminary design report for a new vinyl chloride monomer (VC) production facility to be built in Texas, USA. The planned plant will have a production capacity of 100,000 lb/hr of VC and will supply	The VC plant produces 100,000 lb/hr (45,359 kg/hr) via **oxychlorination-EDC pyrolysis route**: **Major Unit Operations:** • **Oxychlorination Reactor**: Ethylene + HCl + O_2 → EDC (1,2-dichloroethane) • **Direct Chlorination**: Ethylene + Cl_2 → EDC (balanced process) • **EDC Purification Column (C-101)**: Removes lights (chloroethanes) and heavies (trichloroethanes) • **EDC Pyrolysis Furnace**: EDC → VC + HCl at 500°C, ~50% conversion • **Quench System**: Rapid cooling to prevent polymerization • **VC Purification Column (C-102)**: Separates VC from unreacted EDC and HCl • **HCl Recovery**: Recycled to oxychlorination **2. Distillation Column Simulations**

		downstream polyvinyl chloride (PVC) manufacturing operations. Your task is to prepare a professional introductory design report that demonstrates your understanding of the process, simulates key separation units, and critically evaluates how AI tools can support — and fail — in process design decision making. Your work must address the following:Your	**Column C-101: EDC Purification** **Feed Conditions:** • Flow: 100,000 kg/hr crude EDC • Composition: EDC (96.5 mol%), chloroethane (2%), trichloroethanes (1.5%) • T = 80°C, P = 2 bar **DSTWU (Shortcut) Results:** • Stages: 18 theoretical • Feed stage: 9 • Reflux ratio: 2.1 • Distillate purity (EDC): 99.8 mol% • Bottoms purity (heavies): 98.5 mol% • Reboiler duty: 4.2 MW • Condenser duty: -3.8 MW **RadFrac (Rigorous) Results:**

<table>
<tr>
<td></td>
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<td>work must address the following:
Simulate two distillation columns.
Summarize major unit operations: pyrolysis, separation etc. For each column, simulate with Shortcut model (DSTWU) and Rigorous (Radfrac) model. Include: Feed composition, temperature, and pressure; Column configuration (number of stages, feed stage location); Reflux ratio and operating pressure;</td>
<td>• Actual stages: 22 (20% tray efficiency factor)
• Feed stage: 11
• Optimized reflux: 2.4
• EDC recovery: 99.9%
• Reboiler duty: 4.6 MW (+9.5% vs shortcut)
• Condenser duty: -4.1 MW
• Pressure drop: 0.15 bar
Key Design Considerations:
• Operating pressure (1.5-3 bar): Higher P enables cooling water condensers but increases reboiler temperature. Optimal at 2 bar.
• Tray efficiency: Sieve trays estimated at 75-80% for chlorinated hydrocarbons
• Materials: Stainless steel 316 for corrosion resistance
Column C-102: VC/EDC Separation
Feed Conditions (Post-Pyrolysis):</td>
</tr>
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		Distillate and bottoms purity; Energy consumption and condenser/reboiler duties. Discuss the key design considerations influencing column operation. Highlight trade-offs encountered. Graphically analyze effects of pressure, feed stage, reflux ratio, and the total number of stages. Suggest distillation column design parameters (stage count, reflux ratio, operating pressure) • Propose operating	• Flow: 95,000 kg/hr • Composition: VC (52 mol%), EDC (45 mol%), HCl (3 mol%) • T = 40°C, P = 8 bar (compressed to prevent vapor feed) **DSTWU Results:** • Stages: 35 theoretical • Feed stage: 18 • Reflux ratio: 3.8 • VC purity (distillate): 99.95 mol% • EDC recovery (bottoms): 99.5% • Reboiler duty: 12.5 MW • Condenser duty: -11.8 MW **RadFrac Results:** • Actual stages: 45 • Feed stage: 23

		conditions or alternative column configurations • Suggest optimizations for energy integration or column sequencing	• Optimized reflux: 4.2
			• VC product spec: 99.98% (polymerization-grade)
			  ○ Reboiler duty: 13.8 MW (+10.4%)
			  ○ Condenser duty: -12.9 MW
			  ○ Column pressure: 6-7 bar (refrigerated condenser at -10°C)
			Critical Design Factors:
			• **VC volatility**: Boiling point -13.4°C requires refrigeration or elevated pressure
			• **Polymerization risk**: Inhibitor (phenol, 50 ppm) added; residence time <2 hours
			• **Relative volatility (α = 2.8)**: Moderate separation difficulty
			• **HCl removal**: Caustic wash before C-102 to prevent corrosion
			3. Parametric Analysis
			Effect of Operating Pressure (C-102)

			Pressure (bar) / **Condenser T (°C)** / **Reboiler Duty (MW)** / **Utility Cost**

Pressure (bar)	Condenser T (°C)	Reboiler Duty (MW)	Utility Cost
5	-20	13.2	High (refrigeration)
7	**-5**	**13.8**	**Moderate**
10	+15	14.9	Low (cooling water)

Trade-off: P = 7 bar balances refrigeration cost vs. excessive reboiler duty.

Feed Stage Optimization (C-102)

Rigorous simulation showed optimal feed at stage 23/45 (51% from bottom). ±3 stages increases reboiler duty by 8-12%.

Reflux Ratio Sensitivity (C-102)

- R = 3.0: VC purity drops to 99.1% (unacceptable)
- R = 4.2: Meets spec (99.98%)
- R = 5.5: Only 1.5% duty reduction (diminishing returns)

Stage Count Impact

			Reducing C-102 from 45 to 35 stages requires R = 6.1 (18% energy penalty). **Recommended: 45 stages with R = 4.2.** **4. Design Recommendations** **C-101 (EDC Purification):** • 22 sieve trays, feed at stage 11 • R = 2.4, P = 2 bar • Kettle reboiler (LP steam), cooling water condenser **C-102 (VC/EDC Separation):** • 45 valve trays (better turndown for polymerization control) • Feed at stage 23, R = 4.2, P = 7 bar • Refrigerated condenser (ammonia, -5°C) • Forced-circulation reboiler (low ΔT, reduced fouling) **Energy Integration:** • EDC bottoms (120°C) preheat pyrolysis feed: **Save 1.8 MW**

			• VC overhead vapor sub-cool against EDC feed: **Save 0.9 MW** • C-102 overhead compression (7→8 bar) before condensation **Alternative Configuration:** Divided-wall column (DWC) combining C-101 light-end removal with VC/EDC split could reduce capital by 25% and energy by 15%, but increases control complexity. **5. AI Tools in Process Design: Critical Evaluation** **Where AI Excels:** • **Property estimation**: UNIFAC/COSMO-based models predict VLE with <5% error for chlorinated systems • **Optimization loops**: AI-driven sensitivity analysis (tested 200+ cases in minutes) • **Pattern recognition**: Flagged potential azeotrope at EDC/trichloroethane (verified in literature) **Where AI Fails:**

			• **Safety-critical decisions**: AI suggested $P = 12$ bar for C-102 (ignored polymerization runaway risk at $T > 60°C$) • **Material selection**: Recommended carbon steel (catastrophic — HCl requires SS316/Hastelloy) • **Operability**: Proposed $R = 2.8$ without startup/turndown analysis (infeasible at 70% load) • **Regulatory context**: Did not flag Texas CEQ emission limits for chlorinated VOCs **Human Oversight Essential For:** • Process hazard analysis (VC is carcinogenic, explosive) • Mechanical design codes (ASME Sec. VIII for pressure vessels) • Vendor data integration (tray/packing manufacturers) • Economic trade-offs (CAPEX vs. 20-year NPV) **6. Conclusion** The preliminary design delivers 100,000 lb/hr VC with 99.98% purity via two optimized columns. C-101 (22 stages, R=2.4) and C-102 (45 stages, R=4.2) consume 18.4 MW total with 2.7 MW

			recoverable through heat integration. Operating pressure selection (7 bar for C-102) balances refrigeration cost against thermodynamic efficiency. AI tools accelerated simulation but required engineer validation for safety, materials, and operability. Next phase: detailed hydraulic design, control philosophy, and HAZOP study.

6. **Comparison Table**

Table 11: Comparison of AI Output and Team Solved Solution		
Topic	AI	Team
Operating Conditions	ChatGPT: **Column A:** feed ~35 °C at 6–8 bar; operate ~8.5 bar **Column B:** ~25–35 °C, 9 bar	The process consists of three main steps: direct chlorination, pyrolysis, and distillation. The HCl removal column (C-100) runs at 9–14 bar g (≈12 atm abs) with a cold brine condenser to remove hydrogen chloride overhead. The VCM/EDC column (C-101) operates at ~5 atm abs, separating VCM overhead at 35–40 °C and EDC bottoms near 140–150 °C. Temperature and pressure are controlled by valves (VLV-100, 101) and external heat exchangers.
	Gemini: Gives typical plant temps/pressures; no column-specific numbers.	
	Claude:	

	Column A: fed at 80 °C and 2 bar **Column B:** fed at 40 °C and runs around 6–7 bar with a cold condenser.	
	Perplexity: VCM product column ~1.3–2.0 bar; feed 40–60 °C.	
	Mistral: VCM column 5–8 bar.	
Purity	ChatGPT: VCM product about 99.95–99.98%	VCM product purity is 99.98 % (polymer-grade). The HCl column delivers nearly pure HCl overhead, while the VCM/EDC column recovers more than 99% EDC in the bottoms for recycling.
	Gemini: VCM ≥99.9%	
	Claude: VCM 99.98%	
	Perplexity: VCM ≥99.99%	
	Mistral: VCM >99.9%.	
Energy Use	ChatGPT: **Column A:**	D-101 reboiler = 1.46×10^6 cal/s (6.1 MW), D-102 reboiler = 9.98×10^5 cal/s (4.18 MW). Total ≈ 10.3 MW

	Qc 20–23 MMBtu/h Qr 23–26 MMBtu/h; **Column B:** Qc 3–4.5 MMBtu/h. Gemini: No numbers. Claude: **Column A:** Qr 4.6 MW Qc –4.1 MW **Column B:** Qr 13.8 MW Qc –12.9 MW. Perplexity: 2–4 GJ/ton VCM Mistral: No numbers.	for both columns. Condensers remove ≈ –9.8 MW combined. Energy reduction via heat integration with EDC bottoms/furnace feed
Reflux Ratio	ChatGPT: **Column A**: 1.5–1.7 **Column B:** 1.3–1.5 Gemini: 1.0–3.0	RadFrac shows D-101 reflux = 0.314, D-102 reflux = 0.14–0.13. When converted to operating basis (total vs. Rmin ≈ 0.27–0.12), both are within expected moderate-to-low ranges; stable reflux achieved at low ratios due to favorable volatilities.

	Claude: **Column A:** 2.4 **Column B:** 4.2 Perplexity: 2–5 Mistral: 2–5	
Stage Count	ChatGPT: **Column A:** 52 **Column B:** 32 Gemini: **VCM:** 40–60 **HCl:** 30–50 **EDC:** 20–40. Claude: C-101 = 22 actual (≈18 theoretical); C-102 = 45 actual. Perplexity: 30–50 theoretical Mistral: **VCM:** 40–60 **HCl:** 30–50	The HCl column has about 25–30 theoretical stages, sufficient for sharp acid–organic separation. The VCM/EDC column has ≈ 45 stages (tray efficiency ~70 %), with feed introduced near mid-height for best performance.

	EDC: 20–40	
Operating Pressure	ChatGPT: **Column A:** 8.5 bar **Column B:** 9 bar Gemini: 1.3–2 bar Claude: **Column A:** 2 bar **Column B:** 6–7 bar Perplexity: 1.3–2 bar Mistral: **VCM:** 5–8 barg **EDC:** recycle 2–5 barg **HCl:** 5–10 barg	Pressures are set by safety and condensation needs: HCl column ≈ 12 atm abs (9–14 bar g) to condense HCl with brine; VCM/EDC column ≈ 5 atm abs, high enough to condense VCM with cooling water but low enough to avoid decomposition.

7. Reflection Summary

The goal of this project was to evaluate how artificial intelligence (AI) tools can assist in the preliminary design of a vinyl chloride monomer (VCM) production facility. Specifically, there are two distillation columns within the process, the hydrogen chloride (HCl) removal column and the VCM/ethylene dichloride (EDC) separation column. Multiple AI models, including ChatGPT, Gemini, Claude, Perplexity, and Mistral, were prompted to perform the same engineering task, and their results were compared to the team's manually validated Aspen PLUS simulations and design ideas.

Through this comparison, it became clear that each AI system approached the problem differently, reflecting variations in their training data, reasoning styles, and technical depth. While all tools provided accurate general chemical engineering insights, they were less accurate in terms of thermodynamic reasoning and handling process-specific details. This reflection overall aims to summarize how each model performed, identify where AI tools excelled and failed, and discuss the engineering judgment needed to correct their outputs. Ultimately, it highlights how this process has altered the understanding of how AI can support chemical engineers in real-world design applications.

One of the most interesting outcomes of this study was observing how each AI model interpreted the same question in unique ways. Each tool applies its own internal logic and level of technical depth when suggesting operating conditions, purities, energy consumption, reflux ratios, stage counts, and pressures.

ChatGPT took a highly structured and analytical approach. It treated the prompt as a professional design problem and organized its response according to each parameter requested. ChatGPT's output resembled an engineering design brief, clear, logically formatted, and relatively consistent with textbook distillation design values. It provided column pressures, stage counts, and reflux ratios with realistic ranges and included brief justifications for each choice (for example, operating the first column near 8.5 bar to allow cooling-water condensation). ChatGPT demonstrated a good balance between theoretical understanding and practical reasoning, showing a clear awareness of thermodynamic trade-offs, such as condensation temperature and polymerization control. However, it relied heavily on typical data rather than rigorous calculations.

Gemini, by contrast, provided more general industrial information without detailed numbers. It discussed the general process of VCM production (oxychlorination and EDC pyrolysis) but did not specify actual operating parameters for the distillation columns. Its strength lay in contextual breadth — it captured the big picture well — but it lacked quantitative specificity. Gemini's results would be most useful for someone at the conceptual stage of design but insufficient for simulation validation.

Claude produced the most technically specific answers among the models, giving clear numerical outputs for each parameter. It provided feed conditions, operating temperatures, and pressures that aligned closely with industrial references, such as EDC feed at 80 °C and 2 bar, and the VCM/EDC column operating near 6–7 bar with refrigeration. Claude's structured outputs (including column duties and feed stage information) mirrored an engineer's summary of Aspen Plus results. However, while Claude captured numerical precision, it occasionally introduced inconsistent assumptions, such as using unrealistic condenser or reboiler duties in megawatts that exceeded typical scale expectations. These inconsistencies hinted that AI could approximate magnitudes but lacked thermodynamic validation.

Perplexity focused on summarizing best-practice design ranges rather than process-specific data. It provided generic ranges for purity (≥99.99% VCM), reflux ratio (2–5), and operating pressure (1.3–2 bar). Its values were reasonable but lacked any sense of process context. The model seemed to rely more on encyclopedic knowledge than process logic — for instance, suggesting a very low column pressure (1.3 bar) that would not be practical for condensing hydrogen chloride, revealing a gap in thermodynamic understanding.

Mistral provided a hybrid answer — broad but somewhat aligned with industry standards. It offered pressure and reflux ratio ranges in line with industrial norms (VCM columns: 5–8 bar, EDC: 2–5 bar) and mentioned standard unit operations, such as EDC cracking and separation. However, it did not justify its recommendations or demonstrate sensitivity to process safety (e.g., polymerization risks or brine refrigeration). Mistral's approach was more descriptive than analytical, making it weaker for detailed design verification.

Overall, the AI models demonstrated different design personalities. The team's simulation, however, demonstrated that none of the models could independently reproduce accurate design data without human validation and process modeling.

Despite their limitations, each AI model brought unique strengths that accelerated the engineering workflow.

Speed and breadth of information were the most obvious advantages. AI tools provided instant access to literature-like process descriptions that would normally require hours of manual research. For example, ChatGPT's response not only outlined typical operating ranges for each column but also described realistic trade-offs between reboiler duty, pressure, and condenser cooling requirements. This ability to synthesize multiple sources into coherent narratives made it particularly useful during the early stages of design brainstorming.

Creativity and process visualization were other strengths. The AI models, especially ChatGPT and Claude, can construct block flow diagrams (BFDs) and process flow diagrams (PFDs) with clear, step-by-step sequences. For instance, ChatGPT's description of EDC cracking followed by quench, HCl removal, and VCM purification matched the logical sequence used in industrial VCM plants. This kind of AI-assisted visualization can help engineers quickly communicate process ideas without needing to draft detailed diagrams from scratch.

Language clarity was another major strength. AI responses were often written in clear, professional engineering language suitable for reports, making them helpful for drafting early design documentation. ChatGPT's output could be directly used to form the background or process overview section of a report with minimal editing.

Claude's precision was valuable for approximating quantitative relationships. Although some of its numbers required correction, it correctly recognized that the two columns in question (HCl removal and VCM/EDC separation) operate at very different pressures due to volatility differences. Its recognition of such patterns suggested strong pattern learning from process data.

Finally, AI's speed in performing qualitative comparisons was unmatched. While running rigorous Aspen simulations required several iterations, AI tools provided immediate estimates for key variables, such as reflux ratio and stage count. This speed allowed the team to establish initial guesses for Aspen input values more efficiently.

While AI tools demonstrate helpfulness and speed, their technical limitations became clear when numerical accuracy, physical reasoning, and consistency were required.

A major weakness was thermodynamic reasoning. None of the AI systems directly accounted for vapor-liquid equilibrium (VLE), Antoine constants, or relative volatility, all of which are core concepts for distillation design. As a result, while their recommended temperatures and pressures sounded reasonable, they often violated physical constraints. For example, Perplexity's suggestion that the VCM product column could operate at 1.3 bar ignored that hydrogen chloride cannot be condensed under those conditions without refrigeration far below 0 °C. Similarly, Mistral's range of 5–8 bars for all columns overlooks the fact that each stream's volatility and corrosion characteristics dictate specific pressures.

Another recurring issue was unit inconsistency. Some models mixed kJ, MW, and MMBtu/h interchangeably or quoted reboiler duties without specifying the process scale. For instance, Claude listed a 13.8 MW reboiler duty for one column and 4.6 MW for another, which is plausible but unexplained, leaving uncertainty about flowrate basis or product purity assumptions. This highlighted a lack of dimensional engineering awareness, and the models could retrieve realistic magnitudes but were unable to confirm physical consistency.

Assumptions were also oversimplified. Gemini, for instance, overlooked the multi-step nature of the process and failed to distinguish between exothermic chlorination and endothermic cracking reactions. Most models treated the system as a single generic distillation rather than an integrated chemical process. This omission could lead to serious design flaws if engineers were to rely solely on AI without a thorough understanding of the process.

AI models also lacked safety awareness and operability insight. None explicitly mentioned the polymerization hazards of VCM or the corrosive nature of HCl, both of which are critical in industrial design. ChatGPT hinted at inhibitor dosing, but others failed to address metallurgy or corrosion control altogether. These omissions reflect the limitations of general-purpose AI in reasoning about industrial safety, which relies heavily on domain expertise rather than textual data.

Finally, AI systems showed limited reasoning coherence across parameters. When ChatGPT increased column pressure for easier condensation, it failed to adjust reboiler temperature or relative volatility accordingly. This disconnected reasoning revealed that AI can only approximate isolated cause-and-effect relationships, not the interdependent nature of process design.

Correcting AI errors requires substantial engineering judgment grounded in chemical process fundamentals and simulation validation. The team's Aspen Plus results were crucial in confirming accurate operating conditions, pressures, and energy requirements.

The first major correction involved specifying the pressure. While most AI models suggested pressures between 1–9 bar, the team determined through literature and Aspen that the HCl removal column must operate near 9–14 bar g (≈12 atm abs) to condense hydrogen chloride using brine. This pressure range aligns with patent data, ensuring that HCl exists as a liquid rather than a gas. The VCM/EDC column, however, required a much lower pressure of ≈5 atm abs to prevent thermal degradation of VCM. This contrast between the two columns, one high-pressure and one moderate, was essential, and only human judgment could resolve it correctly.

The second correction involved energy balances. The AI-predicted reboiler duties (20–26 MMBtu/h) were unrealistic given the actual feed rate and column size. By applying the equation $Q = mCp\Delta T$ and verifying heat loads in Aspen, the team calculated realistic reboiler duties of 6.1 MW for D-101 and 4.18 MW for D-102, totaling roughly 10.3 MW, which is less than half of ChatGPT's estimate. This correction reinforced the importance of verifying AI outputs through energy and mass balances.

A third correction involved reflux ratios. AI predicted reflux ratios as high as 4–5, but the Aspen DSTWU and RadFrac results showed that stable operation was achieved at much lower values (0.3–0.14). The favorable volatility difference between VCM and EDC reduced the need for high reflux, saving energy. This insight required an understanding of VLE behavior, which AI models were unable to calculate.

The stage counts also required adjustment. While AI models suggested anywhere from 30 to 60 stages, the Aspen results confirmed 26 theoretical stages for the VCM/EDC column and 20 for the HCl column, consistent with practical design limits. Without simulation, AI could not refine these numbers based on feed composition and efficiency assumptions.

Finally, the team applied engineering judgment to assess material compatibility. Since HCl and VCM are highly corrosive and prone to polymerization, stainless steel 316 and inhibitor dosing were recommended, which are precautions that no AI model independently proposed. These safety corrections emphasized that real-world engineering decisions extend beyond thermodynamics to materials, safety, and operability, which are areas that still require human expertise.

This exercise fundamentally changed the team's perspective on how AI can support chemical engineers. Initially, AI was viewed as a tool capable of replacing early-stage calculations. However, after comparing its results to simulation and physical reasoning, the team's understanding shifted: AI is a collaborator, not a designer.

AI's greatest value lies in accelerating conceptual design. It can quickly generate process outlines, estimate parameter ranges, and summarize trade-offs. During the ideation phase, AI enables engineers to transition from a blank page to a structured problem in minutes, rather than hours. For example, ChatGPT's organized process flow outline provided a strong template for the team's report structure, saving time that would otherwise have been spent on formatting and literature review.

However, AI's limitations became clear in numerical accuracy and safety reliability. No model could replicate Aspen Plus thermodynamic predictions or account for phase equilibria, and none performed consistent unit conversions. The experience demonstrated that while AI can suggest plausible engineering values, it cannot guarantee thermodynamic feasibility, mass balance closure, or safe operation, all of which are crucial in process design.

More importantly, this exercise illustrated that AI could enhance, but not replace, engineering judgment. The process of verifying AI outputs against Aspen results deepened the team's understanding of why specific design choices are made. For instance, adjusting pressure and temperature to balance condensation and polymerization constraints reinforced core thermodynamic concepts that students might otherwise overlook. AI provided an opportunity to test understanding by requiring human correction.

Finally, the exercise revealed a broader lesson about the role of AI in real-world chemical engineering. As industrial AI tools continue to evolve, their effectiveness will depend on human oversight. In practice, AI can serve as a rapid design assistant, helping with equipment sizing, cost estimation, and report drafting. Overall engineers will remain responsible for validating every result. In other words, AI can amplify engineering productivity, but it cannot replace engineering accountability.

The comparison of AI outputs and team-verified Aspen results demonstrated both the potential and the limits of AI in chemical process design. Each AI tool offered unique strengths, including ChatGPT's organization, Claude's quantitative precision, Gemini's contextual breadth, Perplexity's conciseness, and Mistral's industrial awareness. However, all lacked thermodynamic rigor and engineering judgment. The team's corrections, grounded in simulation and chemical principles, were essential for producing realistic, safe, and efficient process conditions.

This experience ultimately highlighted that AI can accelerate the early stages of chemical process design by providing structured starting points and facilitating the rapid synthesis of ideas; however, it still relies heavily on human expertise for validation, reasoning, and decision-making. Far from replacing engineers, AI serves as a powerful tool that supports critical thinking, accelerates conceptualization, and enhances creative problem-solving capabilities in the evolving field of chemical engineering.

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